

AKADEMIKA

**RFL -RFT-1G-A
RF CIRCUIT TRAINER KIT**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

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It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

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🕒 Design 🕒 Quality 🕒 Value 🕒 Delivery 🕒 Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that their integrity is checked with respect to the packing list. Also ensure that no visible damage as such appears on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

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The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

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In the warranty period if the service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions' bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS for RFL-RFT-A-1G

RF BJT Amplifier

- Frequency Range : 0.3GHz to 0.8GHz
- Gain S21 : > 10dB
- Reverse Isolation : < - 15 dB

RF Low Noise Amplifier

- Frequency Range : 0.3GHz to 0.8GHz
- Gain S21 : > 10dB
- Noise Figure (NF) : < 1.5dB

RF Power Amplifier

- Frequency Range : 0.3GHz to 1GHz
- Gain S21 : > 12dB
- Maximum Output Power : 30dBm

RF MOSFET Amplifier

- Frequency Range : 0.1GHz to 0.5GHz
- Gain S21 : > 10dB
- Reverse Isolation : < - 15 dB

RF Gain Equalizer

- Frequency : 0.4GHz to 0.8GHz
- Insertion Loss : < - 3 dB
- Load condition : Matched

RF LC Oscillator

- Frequency : 30 MHz (+/- 25 MHz)
- Power Level : > - 30 dBm

RF Phase-locked loop

- Input Frequency : 500 KHz – 5 MHz
- VCO Locked Freq : 1.3 MHz (+/- 500 KHz)

RF Frequency Multiplier

- Input Frequency Range : 10-500MHz
- Output Frequency Range : 20-1000MHz
- Impedance : 50 ohm

RF Lumped LPF Filter

- Cut Frequency : 450 MHz (3dB cut-off +/- 50 MHz)
- Filter order : 7
- Insertion Loss : < - 2 dB
- Return Loss : < -10 dB

RF Lumped HPF Filter

- Frequency : 450 MHz (3dB cut-off +/- 50 MHz)
- Filter order : 7
- Insertion Loss : < - 2 dB
- Return Loss : < -10 dB

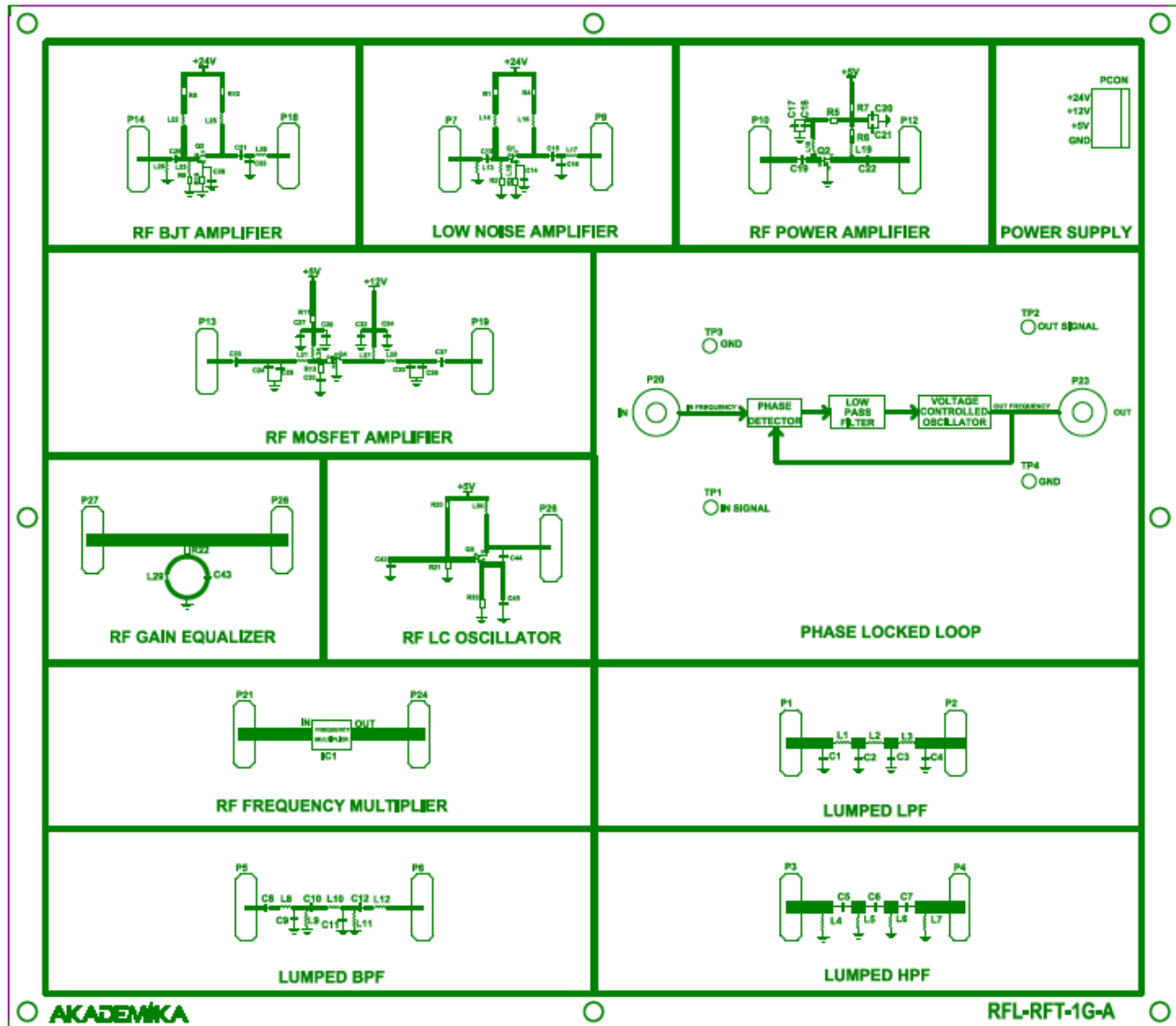
RF Lumped BPF Filter

- Center Frequency : 350 MHz (+/- 70 MHz)
- Filter order : 5
- Bandwidth : ~ 150 MHz @ 3 dB

Total RFL-RFT-1G-A Deliverables

Sr. No	Description	Quantity
1	RFL-RFT-1G-A MODULE BOARD	01
2	DC Power Supply Unit	01
3	SMA (M) to SMA (M) 50 ohm Flexible Cable (50 CM)	03
4	Type N (Male) to SMA (M) 50 ohms Coaxial Cable (1M)	02
5	50 ohm Termination	01
6	BNC to BNC Male Cable	02
7	Experimental manual	01

FUNCTIONAL BLOCK



**EXPERIMENT
NO.1
RF BJT AMPLIFIER**

EXPERIMENT NO: 1

Theory and Experimentation of RF BJT Amplifier

OBJECTIVE

To comprehend the purpose and operational principle of the RF BJT Amplifier by using the RFL-RFT-1G-A module

AIM

To explore the properties of the RF BJT Amplifier module.
Study the RF BJT Amplifier circuit.
Measurement of GAIN of BJT Amplifier.

EQUIPMENTS

RFL-RFT-1G-A KIT, SPECTRUM ANALYZER

THEORY

The RF BJT is a bipolar junction transistor that is designed to operate efficiently at higher RF frequencies in the 100s of MHz into the GHz range. A standard BJT would have unwanted losses in power transfer and may have bad characteristic transistor functionality at those same frequencies.

RF BJT components are used very frequently in power amplification applications. For high frequency signals, like RF, they are designed to have a very good linear response, meaning that the proportionality between input signal and output signal remains quite stable as long as the linear region ratings are followed and maintained. They can also be used as a switching transistor like a MOSFET, operation outside of the linear region and within its saturation mode. In this case, the signal must be translated into a pulse-width modulation and be passed and amplified through the BJT, where the signal is changed back to the original signal.

RF BJTs typically offer some superior characteristics as compared to RF MOSFETs. They maintain a higher linearity in their linear region and are much cheaper for the same functionality. They have much lower output impedances, and have much higher gain potential.

Steps to Design an RF Amplifier

Steps 1 - Determine Design Stages of the RF Amplifier

- Input Matching network: The impedance line connects with the amplifier with a 50-ohm input.
- Amplifiers one or in stages: It depends on your requirements. Usually, it relates to the circuit gain or if the device calls for single or multiple microphones.
- Biasing network: The biasing system supplies the device with voltage or bias.
- Accessories network: The amplifier needs several features and methods for the device to remain stable.
- Output matching network: The impedance line connects with the amplifier with a 50-ohm output.

Steps 2 - Gain Selection Process for RF Amplifier

- The purpose of gain selection is for when the RF signal amplitude is minimal and is not very useful anywhere in the circuit. It should increase to bring up the signal to noise ratio, so it does not decay when the message goes through the course.
- The main task of the gain amplifier is to improve the signal quality by eliminating extra noise. They are ideal for handling shallow signals from the antenna.
- Not all RF amplifiers will have varying gain; some of them provide a fixed gain value, which is controlled by an external resistor. The last type of RF amplifier is known as a variable gain amplifier.
- It allows you to determine the gain, from a wide range. The variable gain amplifier also controlled by an external resistor that can be digitally programmed or a voltage control method by the analogue operation.

RF Amplifier Technical Specifications

To design an RF amplifier, you need to know about the following technical specifications.

1. Frequency range

The frequency range should be fit for any device so that it can use for all kinds of applications. Usually, the span of the frequency is between 100 MHz to 5 GHz.

2. Gain

The gain is measured by what kind of use you have for the device, but the universally accepted value is between 10 to 20 dB. You should have an extensive range of frequency for this gain. Ideally, any amplifier which has a flat increase of around 100 MHz, lower than about 0.2 dB, would work.

3. Input/output impedance

As mentioned before, this set to 50 ohms for both input and output impedance.

4. Noise figure

The higher the device frequency gets, the higher the noise. It will seriously hinder the performance of the amplifier, so the noise figure is an important aspect you should seek. The signal-to-noise ratio should be similar to the signal-to-noise ratio of the amplifier input and an amplifier output, in dB. Ideally, you should have something in the 2dB range, but anything below 3 dB would work perfectly fine.

5. Output power

Consider the maximum possible level when working with a 50-ohm load, with the supply voltage being at the highest point. The output power measured by dBm and should have an ideal range between 12 to 28 dBm.

6. Third-order intercept and 1-dB compression points

These points are indicators that the amplifier you are using for boosting the power is efficient. Most devices would use some kind of broadband modulation scheme, including decent linearity. It will keep the data retention at an optimum level and ensure the broadband usage is at best.

Based on number of stages

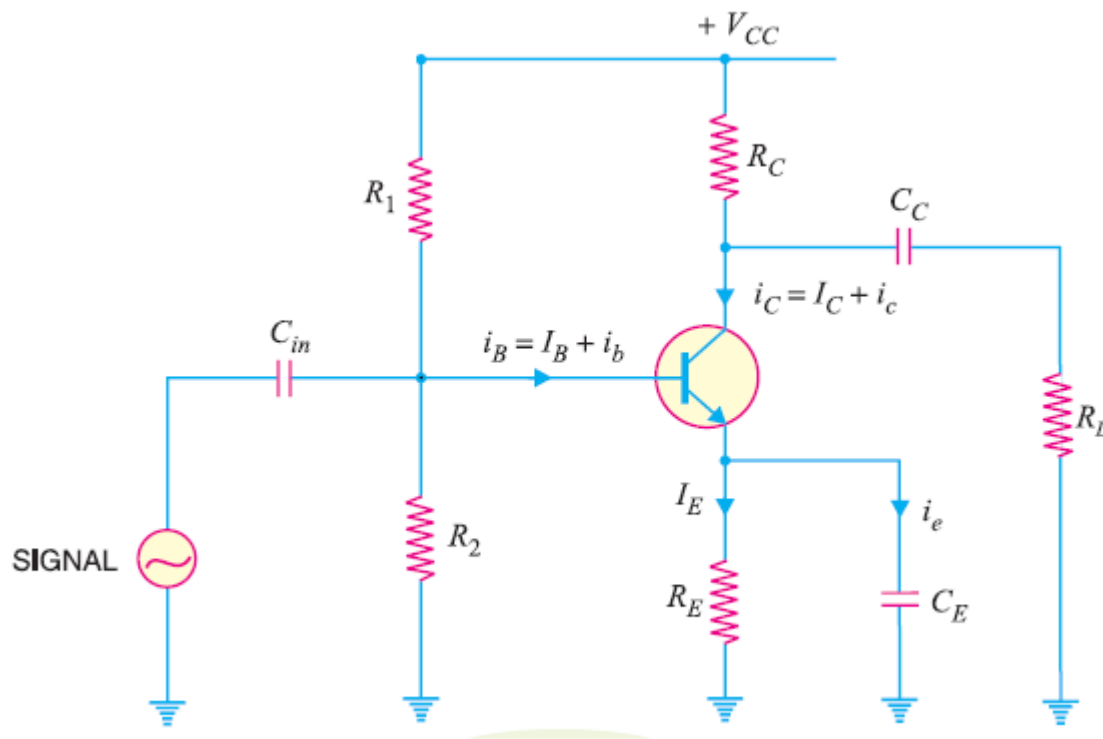
Depending upon the number of stages of Amplification, there are Single-stage amplifiers and Multi-stage amplifiers.

- **Single-stage Amplifiers** – This has only one transistor circuit, which is a single stage amplification.
- **Multi-stage Amplifiers** – This has multiple transistor circuit, which provides multi-stage amplification.

Single Stage Transistor Amplifier

When in an amplifier circuit only one transistor is used for amplifying a weak signal, the circuit is known as single stage amplifier. To achieve faithful amplification in a transistor amplifier, we must use proper associated circuitry with the transistor.

A practical single stage transistor amplifier circuit is shown in fig. below.



The various prominent circuit elements and their functions are as described below.

Biasing Circuit

The resistors R_1 , R_2 and R_E form the biasing and stabilization circuit, which helps in establishing a proper operating point.

Input Capacitor C_{in}

This capacitor couples the input signal to the base of the transistor. The input capacitor C_{in} allows AC signal, but isolates the signal source from R_2 . If this capacitor is not present, the input signal gets directly applied, which changes the bias at R_2 .

Coupling Capacitor C_C

This capacitor is present at the end of one stage and connects it to the other stage. As it couples two stages it is called as **coupling capacitor**. This capacitor blocks DC of one stage to enter the other but allows AC to pass. Hence it is also called as blocking capacitor.

Due to the presence of coupling capacitor C_C , the output across the resistor R_L is free from the collector's DC voltage. If this is not present, the bias conditions of the next stage will be drastically changed due to the shunting effect of R_C , as it would come in parallel to R_2 of the next stage.

Emitter by-pass capacitor C_E

This capacitor is employed in parallel to the emitter resistor R_E . The amplified AC signal is by passed through this. If this is not present, that signal will pass through R_E which produces a voltage drop across R_E that will feedback the input signal reducing the output voltage.

The Load resistor R_L

The resistance R_L connected at the output is known as **Load resistor**. When a number of stages are used, then R_L represents the input resistance of the next stage.

SET UP & PROCEDURE:

Using Spectrum Analyzer

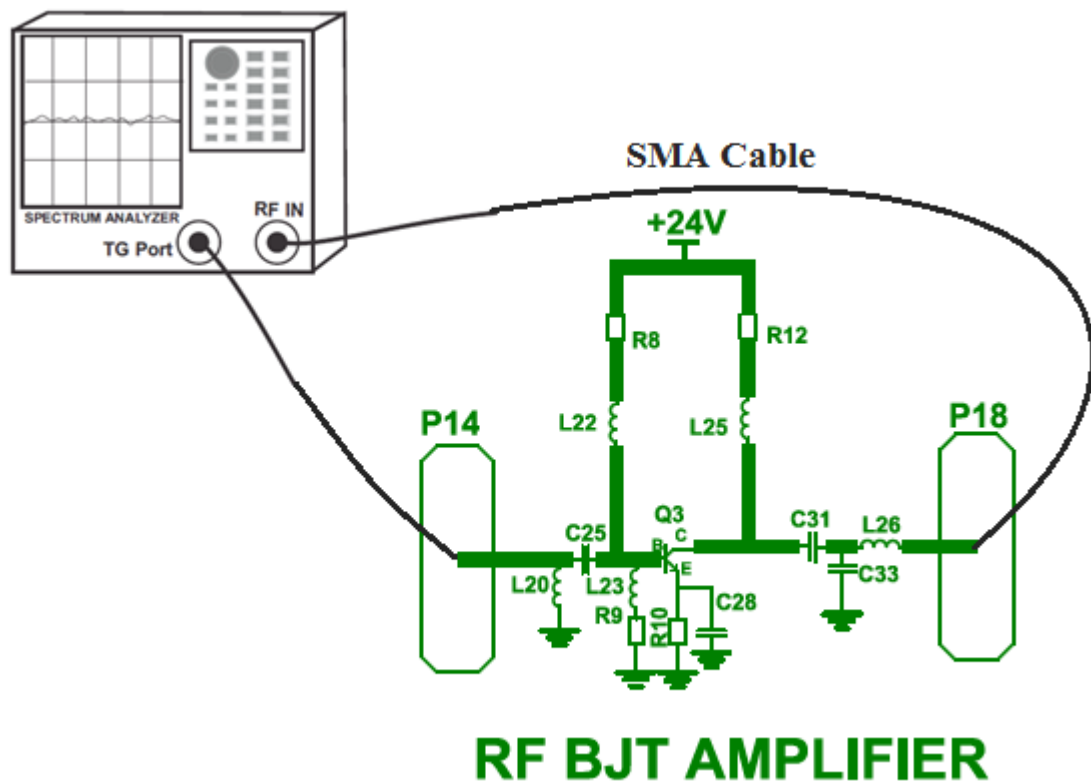
Measurement of Gain (S21) Characteristics of BJT Amplifier

Procedure.

Setting for Spectrum Analyzer:

- Frequency Range : 100 MHz TO 1.0 GHz
- TG Source Power Level : - 25 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.



Setup for Gain (S21) Measurement

Measurement of Gain characteristics (S21):

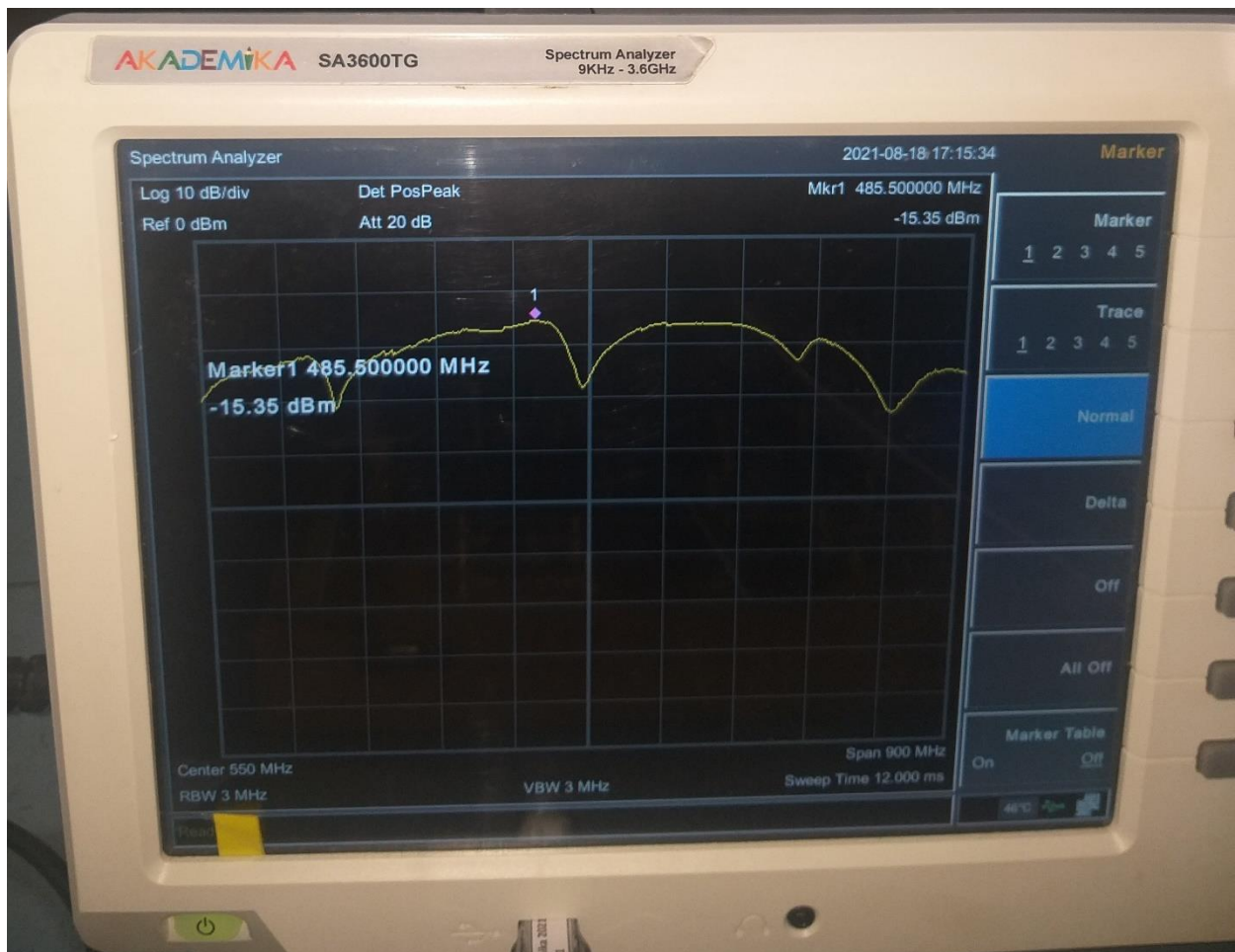
- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the Port 14 of the BJT Amplifier module (i.e. signal of the Spectrum analyzer is applied as input to the BJT Amplifier)
- 2) Connect the Port 18 of the BJT Amplifier module to the Spectrum analyzer Port 2 (i.e. output from the BJT Amplifier is applied to the Spectrum analyzer for data display).
- 3) Connect DC Power Supply to RFT-A Board.
- 4) Turn ON DC Power Supply.
- 5) Apply markers and note down the readings for the frequencies specified in the observation table.
- 6) Observe the output of Gain S21.

Observation Table

Sr.No	Freq (MHz)	Gain (S21) (dB)
1.	400	
2.	485	
3.	650	
4.	700	

Summarize the Gain of BJT Amplifier for the frequencies specified in the observation table.

Reference Plot (S21):



Note : This is the reference graph plot. The graphs vary slightly according to RFT-A boards due to tolerance in PCB and component's.

**EXPERIMENT
NO.2
LOW NOISE
AMPLIFIR**

EXPERIMENT NO: 2

STUDY OF LOW NOISE AMPLIFIER

AIM

To understand the purpose and operational principle of the Low Noise Amplifier (LNA) by using the RFL-RFT-1G-A module

OBJECTIVES:

To explore the properties of RF Low Noise Amplifier.
Gain Measurement of Low Noise Amplifier (LNA).

EQUIPMENTS:

RFL-RFT-1G-A KIT, SPECTRUM ANALYZER.

THEORY

An RF amplifier is an electronic device that is used to amplify a low power radio frequency signal to one with a larger amplitude.

RF amplifiers provide the amplification needed when signals received from an antenna are too small to control the desired device. For instance, the signal picked up by the antenna in a radio receiver is not strong enough to be used in its current form. The RF amplifier increases the signal by enabling it to tune over the desired range of input frequencies.

An RF amplifier is actually a tuned amplifier that enables the input signal of broadcast or transmitted information to control an output signal. The RF amplifier uses frequency-determining networks to convert the input signal into an output signal that will provide the required response at a given frequency. In an RF amplifier, the frequency-determining network is a parallel LC circuit or a tuned circuit. By adjusting the value of inductance or capacitance in the parallel LC circuit, the amplifier is then tuned to the desired frequency. Generally, RF amplifiers are used to amplify signals in any of band of frequencies between 10 kHz and 100,000 MHz. The main characteristics of an RF amplifier are linearity, efficiency, output power, and signal gain. However, RF amplifiers are categorized according to their usage. Below are several categories of RF amplifiers. Click on the category titles for the complete product listings.

Low-Noise Amplifiers (LNAs) – RF amplifiers primarily used in communication systems to amplify weak signals captured by an antenna.

Broadband Amplifiers – RF amplifiers with a flat response over a wide range of frequencies.

RF Power Amplifiers – RF amplifiers that convert a low-power radio frequency signal into a larger signal for driving the antenna of a transmitter.

Solid-State Microwave Amplifiers – RF amplifiers for applications where extensive bandwidth and high gain are necessary.

RF & Microwave Amplifiers – Amplifier used for high-power amplification at low microwave frequencies.

Low-Noise Amplifiers (LNA) :

A low-noise amplifier (LNA) is an electronic amplifier that amplifies a very low-power signal without significantly degrading its signal-to-noise ratio. An amplifier will increase the power of both the signal and the noise present at its input, but the amplifier will also introduce some additional noise. LNAs are designed to minimize that additional noise. Designers can minimize additional noise by choosing low-noise components, operating points, and circuit topologies. Minimizing additional noise must balance with other design goals such as power gain and impedance matching.

Design Consideration of LNA

The block diagram of LNA was given in Fig.1, which consists the source, transistor, *dc* bias, matching network, and the load.

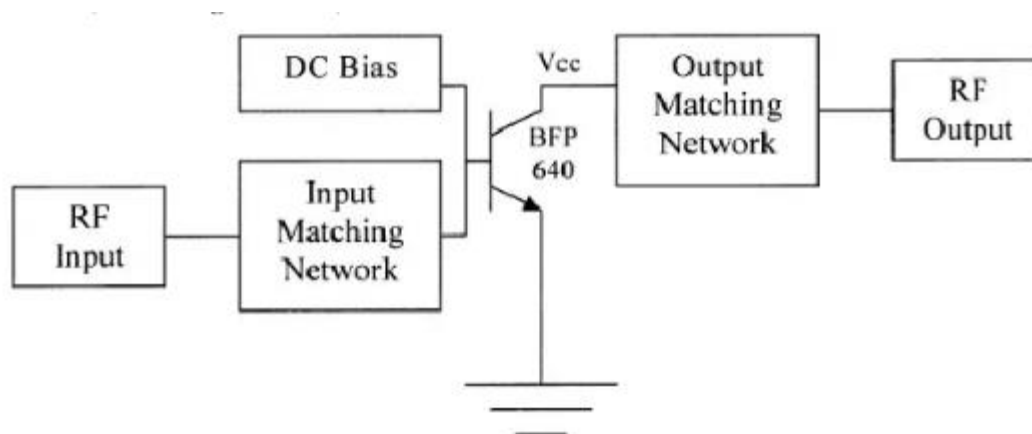


Figure 1. Block diagram of typical LNA design

Low noise amplifiers are the building blocks of communication systems and instruments. The most important LNA specifications or attributes are:

- Gain
- Noise figure
- Linearity
- Maximum RF input

Gain

Amplifiers need a device to provide gain. In the 1940s, that device was a vacuum tube, but now it is usually a transistor. The transistor may be one of many varieties of bipolar transistors or field-effect transistors. Other devices producing gain, such as tunnel diodes, may be used.

Broadly speaking, two categories of transistor models are used in LNA design: Small-signal models use quasi-linear models of noise and large-signal models consider non-linear mixing.

The amount of gain applied is often a compromise. On one hand, high gain makes weak signals strong. On the other hand, high gain means higher level signals, and such high level signals with high gain may exceed the amplifier's dynamic range or cause other types of noise such as harmonic distortion or nonlinear mixing.

Noise figure

The noise figure helps determine the efficiency of a particular LNA. LNA suitability for a particular application is typically based on its noise figure. In general, a low noise figure results in better signal reception.

Impedance

The circuit topology affects input and output impedance. In general, the source impedance is matched to the input impedance because that will maximize the power transfer from the source to the device. If the source impedance is low, then a common base or common gate circuit topology may be appropriate. For a medium source impedance, a common emitter or common source topology may be used. With a high source resistance, a common collector or common drain topology may be appropriate. An input impedance match may not produce the lowest noise figure.

Applications

LNAs are used in communications receivers such as in cellular telephones, GPS receivers, wireless LANs (WiFi), and satellite communications.

In a satellite communications system, the ground station receiving antenna uses an LNA because the received signal is weak since satellites have limited power and therefore use low-power transmitters.

SET UP & PROCEDURE:

Using Spectrum Analyzer

Measurement of Gain (S21) Characteristics of LNA.

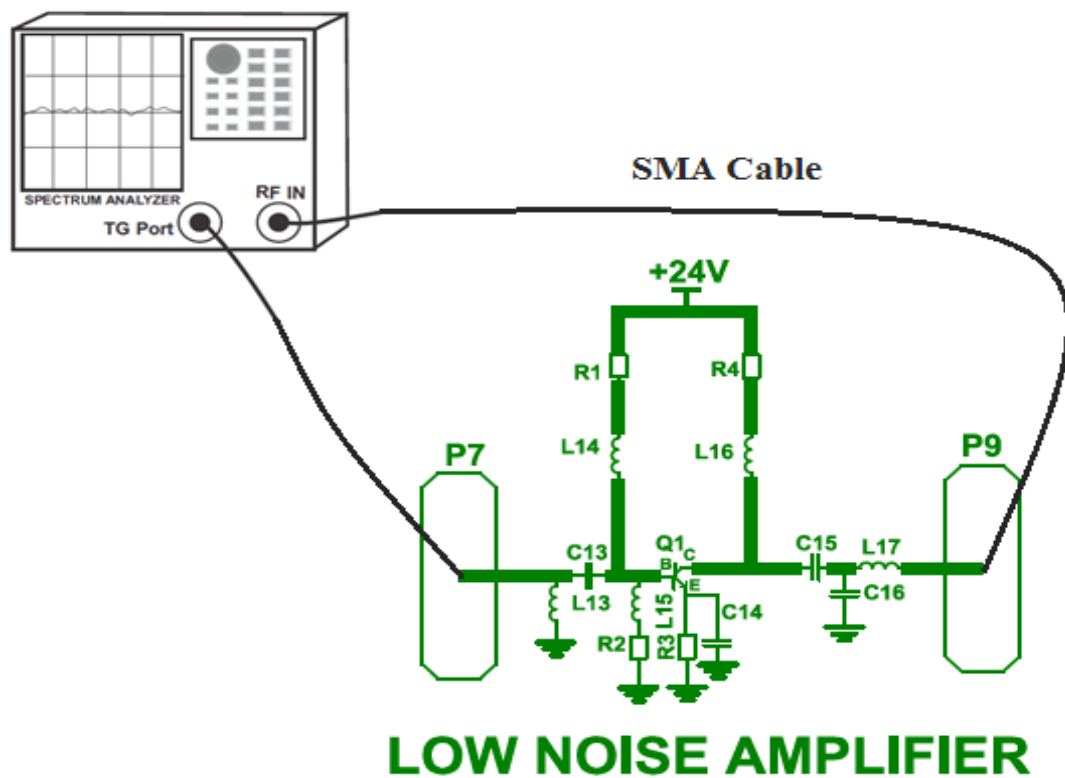
Procedure.

Setting for Spectrum Analyzer:

- Frequency Range : 100 MHz TO 1.0 GHz
- TG Source Power Level : - 25 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.

Measurement of Gain characteristics (S21):



Setup for Gain (S21) Measurement

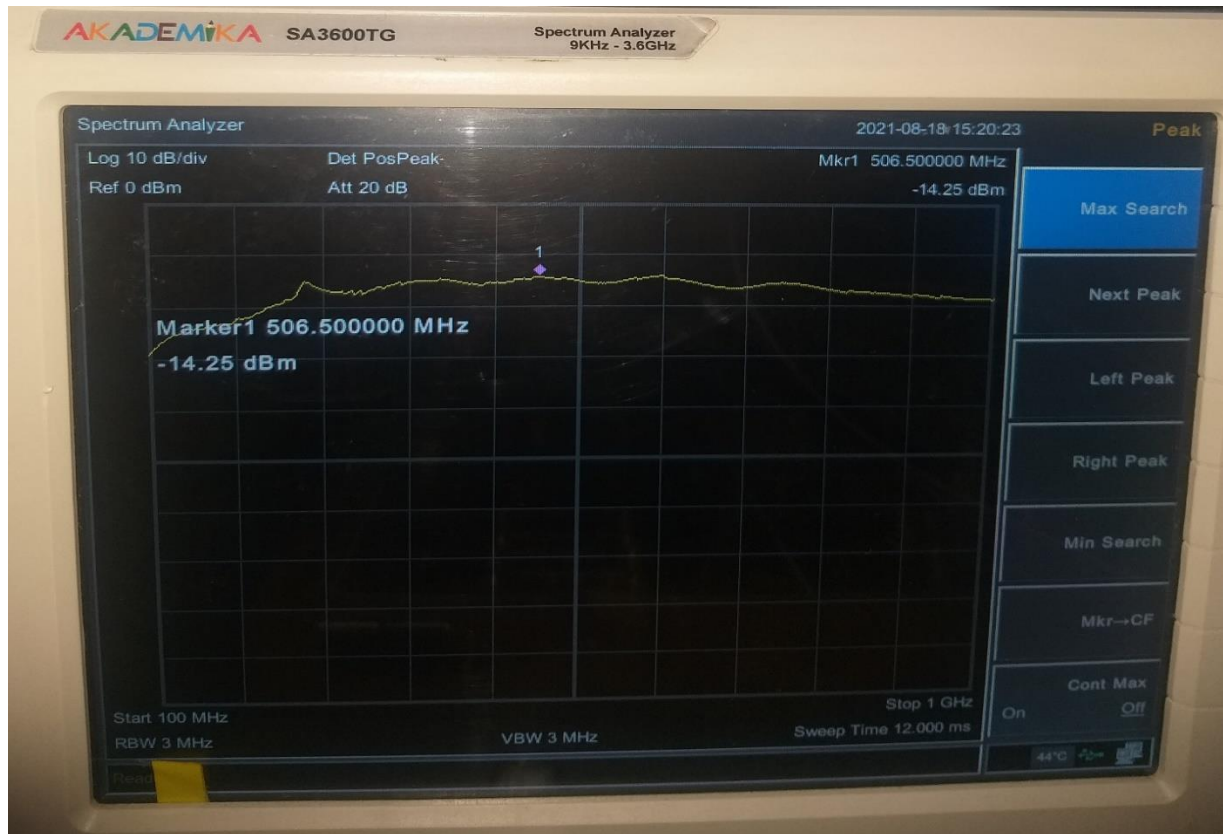
- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to Port P7 of the Low Noise Amplifier module (i.e. signal of the Spectrum analyzer is applied as input to the LNA)
- 2) Connect the Port P9 of the Low Noise Amplifier module to the Spectrum analyzer Port 2 (i.e. output from the LNA is applied to the Spectrum analyzer for data display).
- 3) Connect DC Power Supply to RFT-A Board.
- 4) Turn ON DC Power Supply.
- 5) Apply markers and note down the readings for the frequencies specified in the observation table.
- 6) Observe the output of Gain S21.

Observation Table

Sr.No	Freq (MHz)	Gain (S21) (dB)
1.	400	
2.	500	
3.	600	
4.	700	
5.	800	

Summarize the Gain of LNA for the frequencies specified in the observation table.

Reference Plot (S21):



Note : This is the reference graph plot. The graphs vary slightly according to RFT-A boards due to tolerance in PCB and component's.

**EXPERIMENT
NO.3
RF POWER AMPLIFIR**

EXPERIMENT NO: 3

Theory and Experimentation of RF Power Amplifier

OBJECTIVE

To comprehend the purpose and operational principle of the RF Power Amplifier by using the RFL-RFT-1G-A module

AIM

To explore the properties of the RF Power Amplifier module.

Study the RF Power Amplifier biasing circuit

Gain Measurement of Power Amplifier

EQUIPMENTS

RFL-RFT-1G-A KIT, SPECTRUM ANALYZER

THEORY

A radio frequency power amplifier (RF power amplifier) is a type of electronic amplifier that converts a low-power radio-frequency signal into a higher power signal. Typically, RF power amplifiers drive the antenna of a transmitter. Design goals often include gain, power output, bandwidth, power efficiency, linearity (low signal compression at rated output), input and output impedance matching, and heat dissipation

Many modern RF amplifiers operate in different modes, called “classes”, to help achieve different design goals. Some classes are class A, class AB, class B, class C, which are considered the linear amplifier classes. In these classes the active device is used as a controlled current source. The bias at the input determines the class of the amplifier. A common trade-off in power amplifier design is the trade-off between efficiency and linearity. The previously named classes become more efficient, but less linear, in the order they are listed. Operating the active device as a switch results in higher efficiency, theoretically up to 100%, but lower linearity

Based on its output

Depending upon the parameter that is amplified at the output, there are voltage and power amplifiers.

- **Voltage Amplifiers** – The amplifier circuit that increases the voltage level of the input signal, is called as Voltage amplifier.

- **Power Amplifiers** – The amplifier circuit that increases the power level of the input signal, is called as Power amplifier.

Power Amplifier Requirements

The RF Power Amplifier (PA) is the last component of a transmitter chain. The purpose of a transmitter is to deliver an RF signal with required properties and specified power level to the antenna; and the need for the PA is in amplification of that signal to the level expected at antenna port.

That is, in order to do the job, the PA has to meet the following requirements:

- It has to have sufficient Gain – to amplify RF signal to the level expected at Antenna Port. It is expressed as the difference between input and output RF powers.
- It has to have sufficient Power Handling Capability – to be able to sustain a RF power level expected by the Antenna Port. Sometimes it is expressed as “dissipated power”, however for high power RF transistors it is often given as “Saturated Power” level.
- It has to be distortion-free – in order for a system’s receiver to be able to recognize radiated signal. It is expressed as a degree of linearity.
- It has to be stable – to avoid a creation of oscillations over anticipated variations of external conditions (that is, changes in temperature, load, frequency, DC and RF powers)
- In addition, it is preferable that PA requirements are to be met efficiently. It is needed to avoid wasting of DC power, and/or to preserve the battery drainage. The problem with efficient amplifiers though, is that they do introduce additional distortions which increase with increasing efficiency. In order to deal with this issue, a number of linearization techniques have been developed. Amplifiers are divided into different classes (Class A to C for controlled current source and Class D and above for switched mode amplifiers) and linearization, specific for a given class, is applied.

All these requirements have to be met over the frequency band of operation (which might be 3-to-5 times wider than the occupied bandwidths if linearization techniques are to be used).

A detailed description of requirements is given below.

1. Gain

For a Power Amplifier, gain is defined as the difference between the power of the RF signal applied to PA input and the one delivered to the antenna port. While this property is the most important for the PA (it is the purpose of Power Amplifiers to create Gain), its actual value is of a lesser importance. Really, the gain of a last stage could be easily increased by adding extra preamplifiers preceding the final stage of a Power Amplifier.

That is, the value of the Gain could be reduced for the sake of achieving other parameters (better match, linearity or efficiency). Very often two Gain values are included on PA spec sheets – they are a small signal gain (that is, gain at the significantly reduced level of an input signal; expressed as S_{21}) and a large signal gain (gain at 1 dB compression point).

2. Output Power

Power Handling Capability is defined as the maximum power that an amplifier can handle without damage. Its value depends on the size and configuration of a transistor's die and the proper application of cooling. This parameter should be large enough to sustain an RF power level expected by the Antenna Port. Sometimes it is expressed as “dissipated power” (which is the product of a current flowing through collector/drain and the voltage across the device); however, for high power transistors it is usually given as “output power at 1 dB Compression Point” or “Saturated Power.”

3. Linearity

Linearity is the measure of an amplifier's distortions. Distortions are happening when RF signals with variable envelopes are applied to a nonlinear amplifier; they have to be low enough for the system's receiver to recognize what the transmitter is sending. Mathematically, the distortions are realized as an interaction between additional spectral components created by the amplifier's nonlinear transfer function (I-V DC curves).

For CW signals their measure is Compression Point or Intercept Point; for digitally modulated signals the figures of merit are Adjacent Channel Power (ACP – the measure of out-of-band interference) and Error Vector Magnitude (EVM – the measure of in-channel distortions). Distortions can be compensated by the application of products with the same amplitude and the opposite polarity to undesirable ones – the process which is called “linearization.”

There are two basic approaches to linearization – one of them is called Feedback, where the corrections for creating a distortion-free operation are done at the amplifier's input (predistorters are operating on this principle), and the other is Feed-Forward, where the corrections are applied to the output of the amplifier. The first approach is cheaper, however it cannot compensate for distortions from heavily compressed amplifiers (above 1 dB compression point).

4. Efficiency

Efficiency is a measure of DC energy loss when it is transferred to RF power. In hand-held units, an inefficient PA is draining the battery and producing excessive heat; in high-power stationary units it requires complicated cooling systems and increases the cost of operation.

However having ideal sinusoidal RF signal at an amplifier's output (like in Class A amplifiers) makes the PA lose 50% of its DC power by repeating redundant (positive and negative) information about amplitudes of Voltage and Current; In order to increase

efficiency, the redundancy of Voltage/Current amplitudes has to be eliminated, which is done with the introduction of a Class B amplifier. In this class a conduction angle is reduced by a biasing amplifier to the origin of its transfer function, which makes the PA transfer only half (positive OR negative) of input RF sinusoids. However, due to the nonlinearities of transfer function (especially pronounced at its origin) – maximum efficiency for this method would not exceed 78%.

5. Stability

The RF Power Amplifier has to be stable (that is, oscillation free) over its operational range (over variations in temperature, frequencies, and power levels). Oscillations are caused by a positive feedback from the amplifier's output; one has to be careful to avoid them or to dump them with additional circuitry.

For small signal operations, stable conditions are found from so-called "Stability Factor," which is a formula derived from S-parameters (parameters describing transfer characteristics of a linear circuit). However, for large signal operations (that is, the most important area of Power Amplifier's operation) stability has to be determined from Load Pull measurements.

Wideband Power amplifier design

Impedance transformations over large bandwidth are difficult to realize, thus most wideband amplifiers use 50 Ω output loading. Transistor output power is then limited to

$$P_{out} \leq \frac{(V_{br} - V_k)^2}{8Z_o}$$

V_{br} is defined as the breakdown voltage

V_k is defined as the knee voltage

and Z_o is being chosen so the rated power can be met. The external load is typically $Z_L=50$ ohm. Therefore, there must be some sort of transformation that transforms from Z_o to $Z_L=50$ ohm.

Applications

The basic applications of the RF power amplifier include driving to another high power source, driving a transmitting antenna and exciting microwave cavity resonators. Among these applications, driving transmitter antennas is most well-known. The transmitter–receivers are used not only for voice and data communication but also for weather sensing (in the form of a radar)

SET UP & PROCEDURE:

Using Spectrum Analyzer

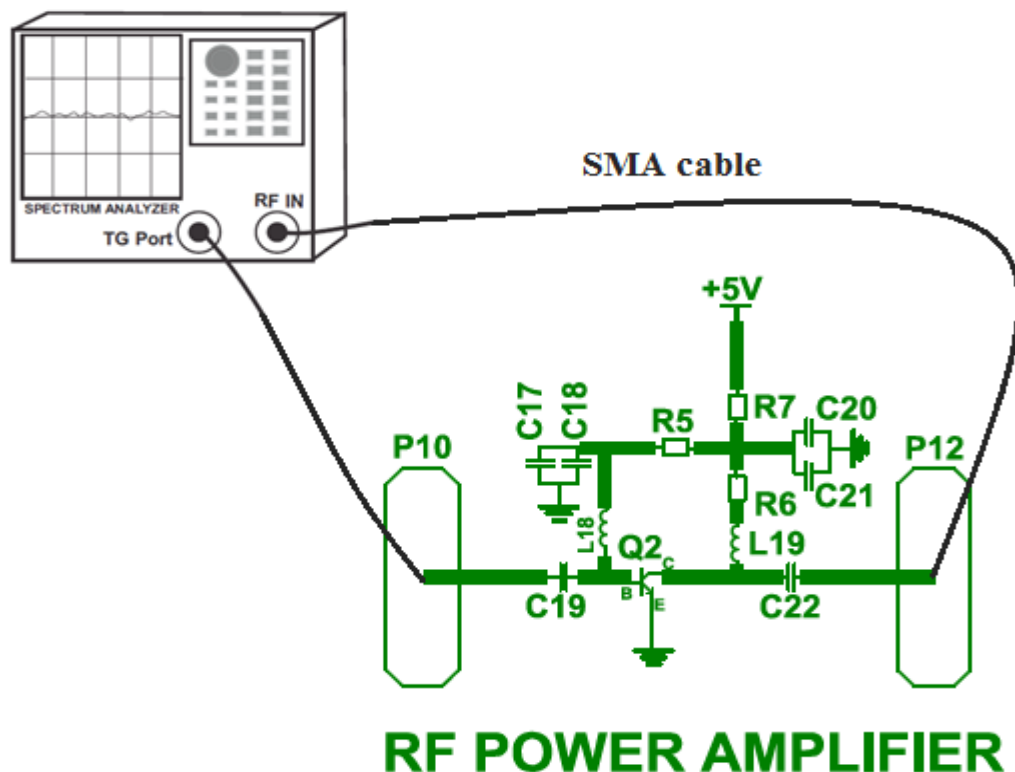
Measurement of Gain (S21) Characteristics of RF Power Amplifier

Setting for Spectrum Analyzer:

- Frequency Range : 100 MHz TO 1.0 GHz
- TG Source Power Level : - 25 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.

Measurement of Gain characteristics (S21):



Setup for Gain (S21) Measurement

Measurement of Gain characteristics (S21):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the Port 10 of the RF Power Amplifier module (i.e. signal of the Spectrum analyzer is applied as input to the Power Amplifier)

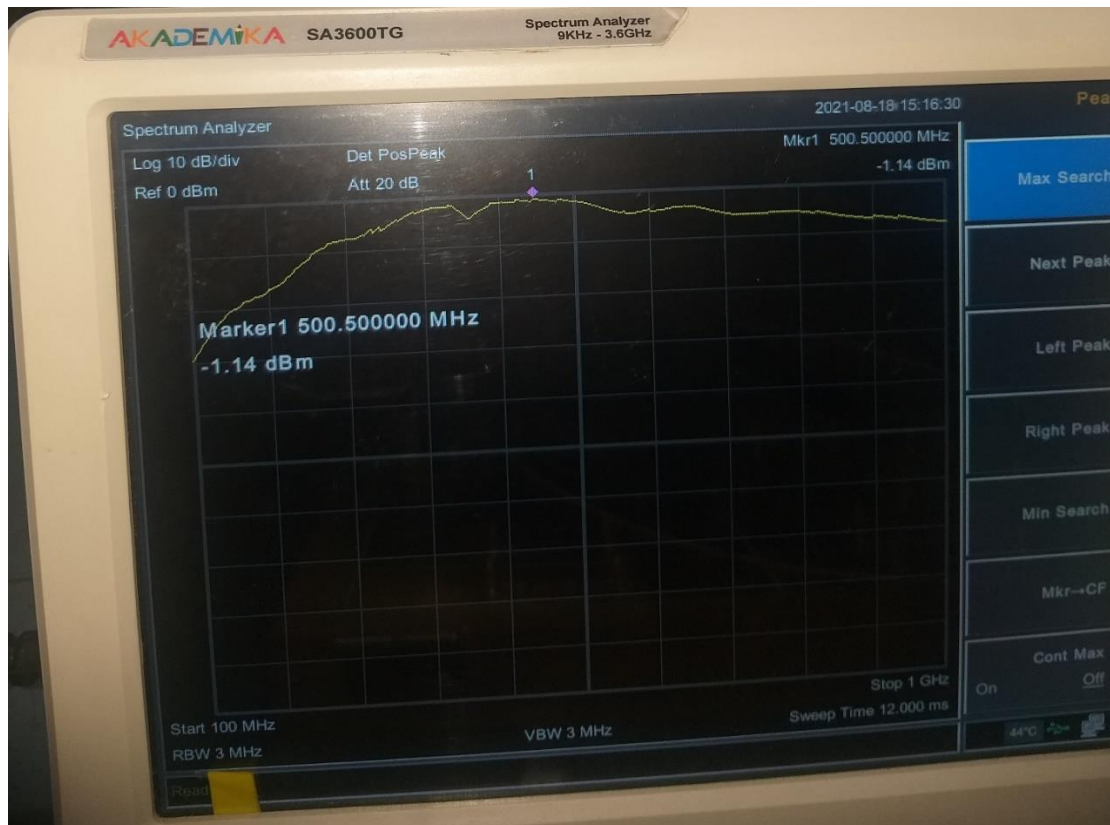
- 2) Connect the Port 12 of the RF Power Amplifier to the Spectrum analyzer Port 2 (i.e. output from the Power amplifier is applied to the Spectrum analyzer for data display).
- 3) Connect DC Power Supply to RFT-A Board.
- 4) Turn ON DC Power Supply.
- 5) Apply markers and note down the readings for the frequencies specified in the observation table.
- 6) Observe the output of Gain S21.

Observation Table

Sr.No	Freq (MHz)	Gain (S21) (dB)
1.	400	
2.	500	
3.	600	
4.	700	
5.	800	

Summarize the Gain of power amplifier for the frequencies specified in the observation table.

Reference Plot (S21):



Note : This is the reference graph plot. The graphs vary slightly according to RFT-A boards due to tolerance in PCB and component's.

**EXPERIMENT
NO.4
RF MOSFET
AMPLIFIER**

EXPERIMENT NO: 4

Theory and Experimentation of RF MOSFET Amplifier

OBJECTIVE

To comprehend the purpose and operational principle of the RF MOSFET Amplifier by using the RFL-RFT-1G-A module

AIM

To explore the properties of the RF MOSFET Amplifier module.

Study the RF MOSFET Amplifier circuit .

Gain Measurement of MOSFET Amplifier.

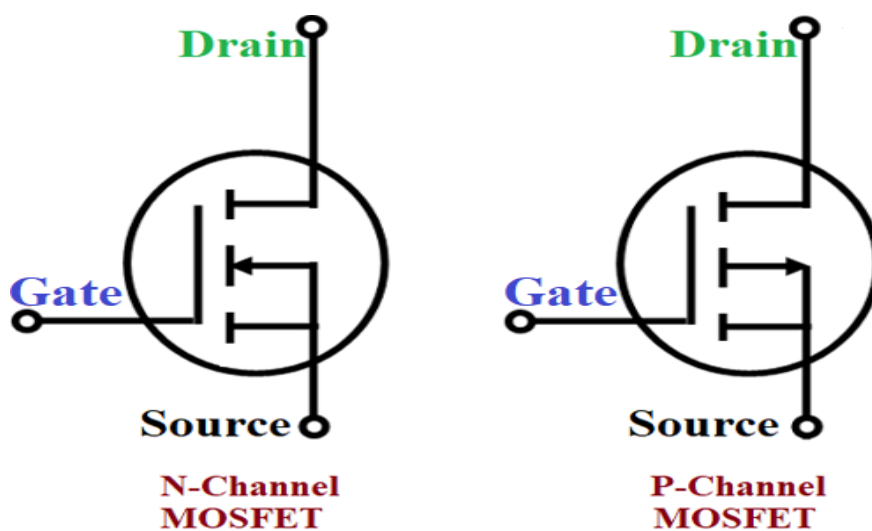
EQUIPMENTS

RFL-RFT-1G-A KIT, SPECTRUM ANALYZER

THEORY

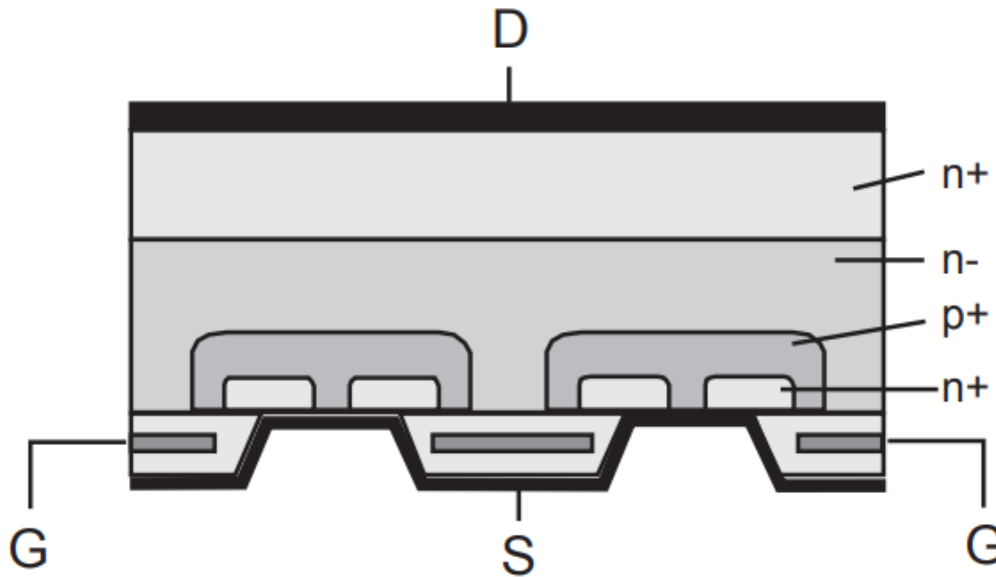
MOSFET stands for Metal Oxide Semiconductor Field Effect Transistor. It's a voltage controlled device with 3 terminals:

- Gate (electrically insulated from the semiconductor)
- Drain
- Source



MOSFET circuit symbol

When a voltage applied between the **Gate** and the **Source** reaches a certain threshold ($V_{GS(th)}$ or *threshold G-S voltage*), the device is able to support current conduction between the **Drain** and the **Source** (I_D or *drain current*).



MOSFET simplified structure

When a voltage applied between the **Gate** and the **Source** is below $V_{GS(th)}$, the device will withstand a voltage up to BV_{DSS} (or *breakdown voltage*).

MOSFETs can be used as a **signal amplifier** (linear operation) or **as a switch** in power applications.

MOSFET parameters

Like many other types of semiconductor power switches, the main parameters of a MOSFET, usually available in most datasheets, are:

- **$R_{DS(on)}$ (on-state resistance)**: electrical resistance when the device is set in on state. The lower is $R_{DS(on)}$, the lower is the conduction loss due to power dissipation when the current is flowing.
- **BV_{DSS} (breakdown voltage)**: maximum drain-to-source voltage that the device is able to sustain when in off state.
- **Q_G (total gate charge)**: amount of electric charge required to the gate driver to turn on/off the device itself. Q_G impacts directly the efficiency (the lower, the better).

The product of $R_{DS(on)}$ and Q_G is known as the **MOSFET Figure of Merit (FOM)**.

Other important parameters are **intrinsic capacitances** that can affect the switching times and voltage spikes, and **body drain diode** when device is used as power diode, like in synchronous free-wheeling operation mode.

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit
$V_{(BR)DSS}$	Drain-source breakdown voltage	$I_D = 250\mu A, V_{GS} = 0$	75			V
I_{DSS}	Zero gate voltage drain current ($V_{GS} = 0$)	$V_{DS} = \text{max ratings}$ $V_{DS} = \text{max ratings},$ $T_C = 125^\circ C$			1 10	μA μA
I_{GSS}	Gate-body leakage current ($V_{DS} = 0$)	$V_{GS} = \pm 20V$			± 100	nA
$V_{GS(th)}$	Gate threshold voltage	$V_{DS} = V_{GS}, I_D = 250\mu A$	2		4	V
$R_{DS(on)}$	Static drain-source on resistance	$V_{GS} = 10V, I_D = 70A$		0.0065	0.0075	Ω

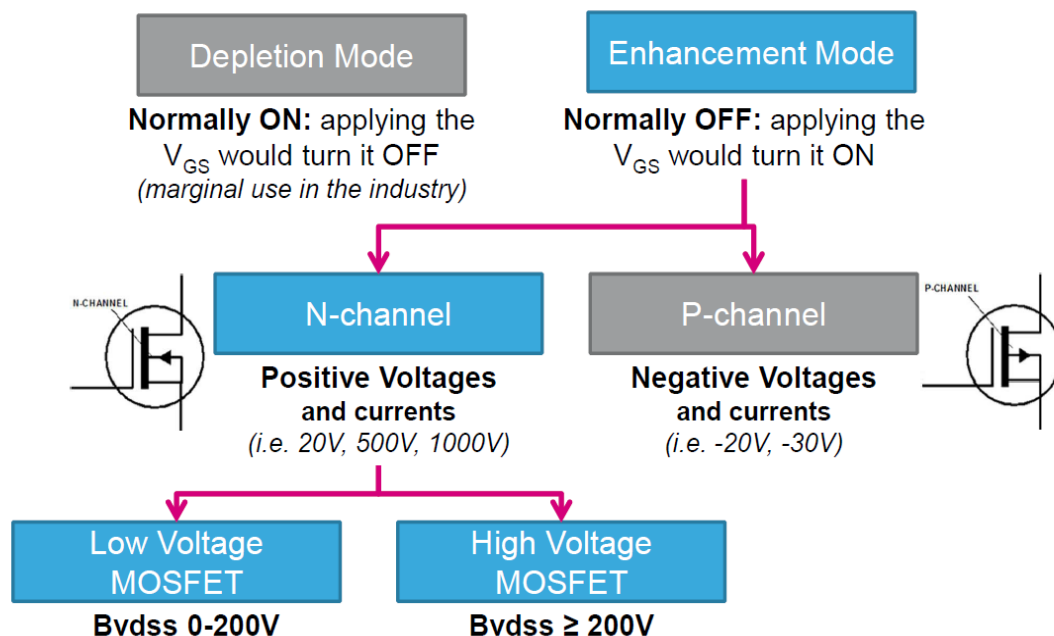
Example of MOSFET datasheet parameters

Main types of power MOSFETs

MOSFETs can be of different types, including:

- Depletion Mode: Normally ON. Applying the V_{GS} would turn it OFF.
- Enhancement Mode: Normally OFF. Applying the V_{GS} would turn it ON.
- N-channel MOSFETs: positive voltages and currents.
- P-channel MOSFETs: negative voltages and currents.
- Low voltage MOSFETs: BV_{DSS} from 0 V to 200 V.
- High voltage MOSFETs: BV_{DSS} greater than 200 V.

N-channel enhancement-mode MOSFETs are the most popular type used in power switching circuits because of their low $R_{DS(on)}$ (on-state resistance) compared to P-channel MOSFETs.



Overview of the main types of MOSFETs

Power MOSFET Applications

Power MOSFETs play an important role in all applications handling Power.

The main applications of **high voltage MOSFETs** include:

- Switch Mode Power Supplies (SMPS)
- Residential, commercial, architectural and street lighting

- DC-DC converters
- Motor control
- Automotive applications

The main applications of **low voltage MOSFETs** include:

- Switch, buck and synchronous rectification
- Uninterruptible Power Supplies (UPS)
- Small motor control
- Switch Mode Power Supplies (SMPS)
- Power-Over-Ethernet (PoE)
- Solar inverters
- Automotive applications

SET UP & PROCEDURE:

Using Spectrum Analyzer

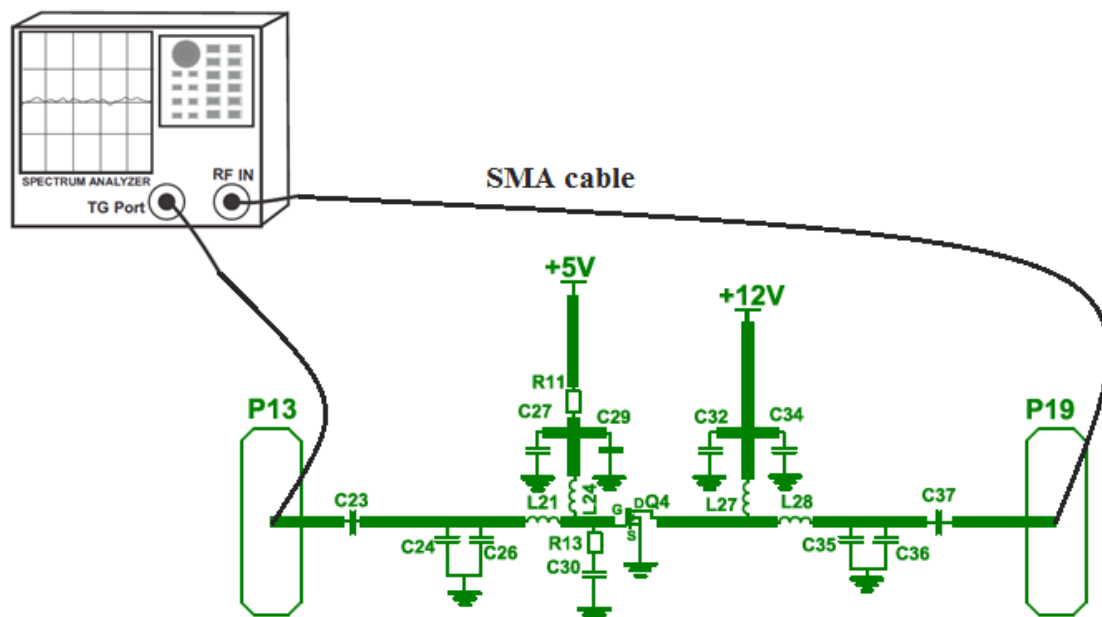
Measurement of Gain (S21) Characteristics of RF MOSFET Amplifier

Setting for Spectrum Analyzer:

- Frequency Range : 100 MHz TO 1.0 GHz
- TG Source Power Level : - 25 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.

Measurement of Gain characteristics (S21):



RF MOSFET AMPLIFIER

Setup Gain S21 Measurement

Measurement of Gain characteristics (S21):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the Port 13 of the RF MOSFET module (i.e. signal of the Spectrum analyzer is applied as input to the MOSFET)
- 2) Connect the Port 19 of the RF MOSFET module to the Spectrum analyzer Port 2 (i.e. output from the MOSFET is applied to the Spectrum analyzer for data display).
- 3) Connect DC Power Supply to RFT-A Board.
- 4) Turn ON DC Power Supply.

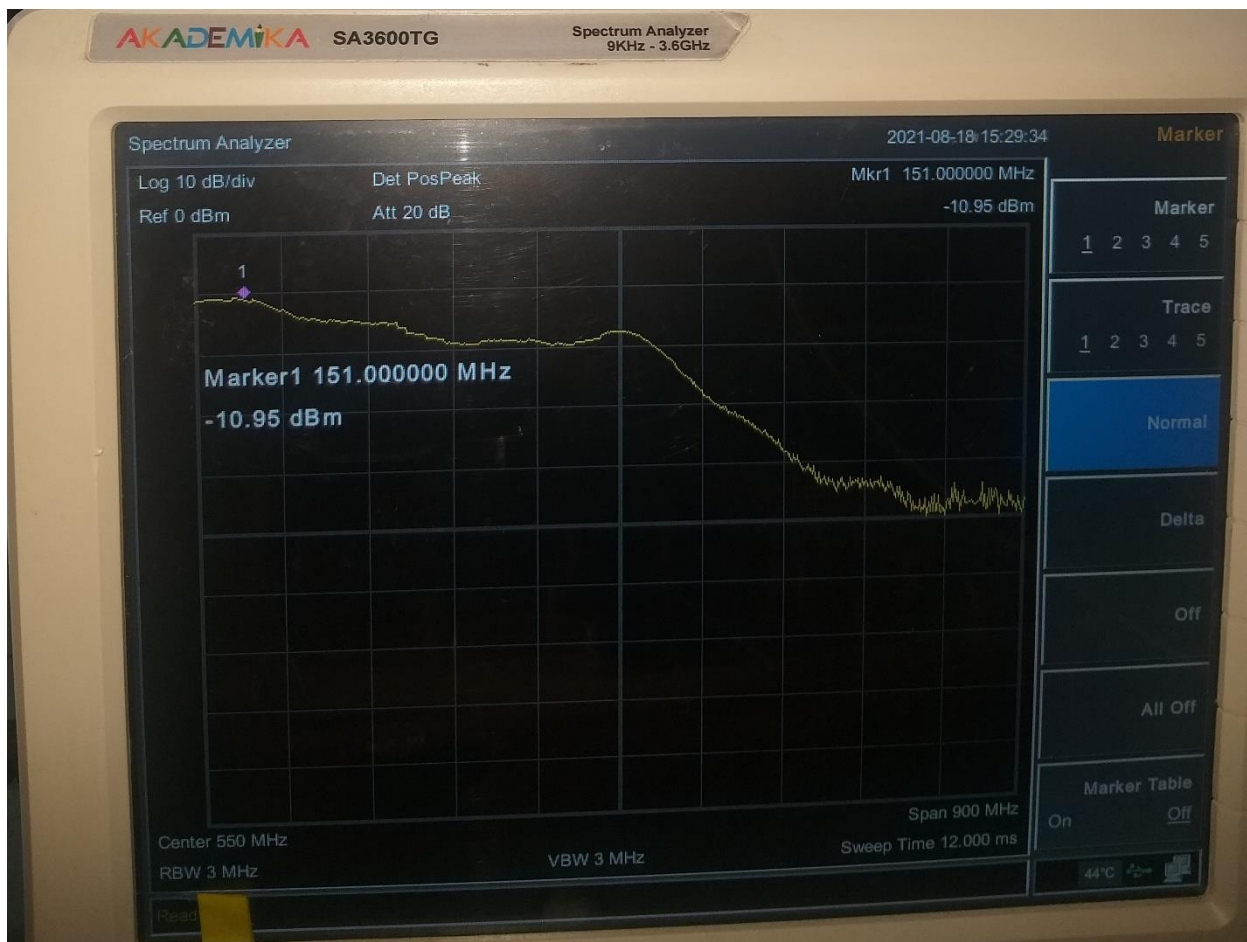
- 5) Apply markers and note down the readings for the frequencies specified in the observation table.
- 6) Observe the output of Gain S21.

Observation Table

Sr.No	Freq (MHz)	Gain (S21) (dB)
1.	100	
2.	150	
3.	250	
4.	350	
5.	500	

Summarize the Gain of MOSFET amplifier for the frequencies specified in the observation table.

Reference Plot (S21):



Note : This is the reference graph plot. The graphs vary slightly according to RFT-A boards due to tolerance in PCB and component's.

**EXPERIMENT
NO.5
RF GAIN EQUALIZER**

EXPERIMENT NO: 5

Theory and Experimentation of RF Gain Equalizer

OBJECTIVE

To comprehend the purpose and operational principle of the RF Gain Equalizer by using the RFL-RFT-1G-A module

AIM

To explore the properties of the RF Gain Equalizer module.

Study the RF Gain Equalizer circuit.

Measurement of GAIN equalization for Power Amplifier.

EQUIPMENTS

RFL-RFT-1G-A KIT, SPECTRUM ANALYZER

THEORY

Equalizers are devices used to compensate for negative gain slope in the frequency response of a wide variety of RF systems. Unlike a standard attenuator with a flat frequency response, an equalizer is a unique kind of attenuator which exhibits lower insertion loss as frequency increases with some known slope. This is a useful characteristic for system designers working in wideband applications where the gain response of circuit elements or of the entire RF chain often varies across frequency.

Gain Equalizers are commonly known as special attenuators that have a frequency response footprint that is intentionally not flat. The RF industry provides many reasons for an equalizer, the most common is to fix wideband problems in microwave systems. In certain instances, RF signals have negative gain slope, (i.e. less gain the higher frequency you go). This includes not only amplifiers, but passive components like coaxial cables and microstrip transmission lines. The antidote to negative slope is adding a linear gain equalizer that adds a positive gain slope to your line-up.

1. Categorizing Gain Equalizers by Power Supply Requirement

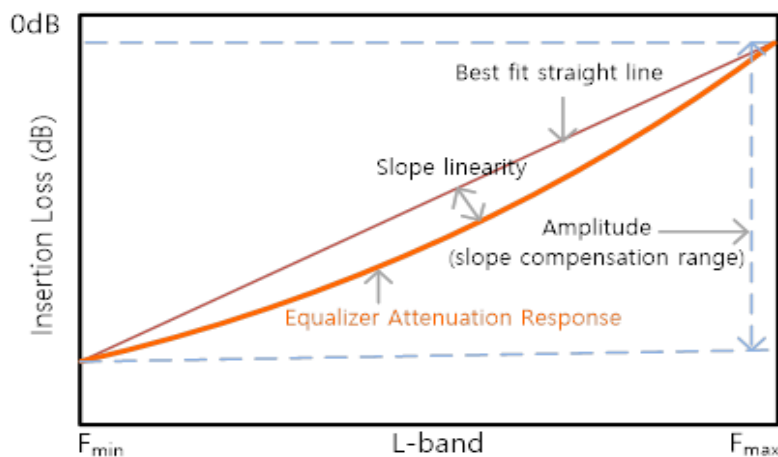
Gain Equalizers can be categorized into Active Gain Equalizers and Passive Gain Equalizers based on whether they require a power supply. For example, Gain Equalizers that require an external power source are called Active Gain Equalizers. Those that do not require an external power supply are called passive gain equalizers.

Active Gain Equalizers produce a positive gain characteristic because they use an amplifier to amplify the signals it receives. On the other hand, Passive Gain Equalizers produce an insertion loss characteristic because they do not use an amplifier.

Type	Advantages	Disadvantages
Passive Gain Equalizer	Low cost and small in size	Only used in systems that can compensate for insertion loss of Gain Equalizers
Active Gain Equalizer	Users can control and tune its positive gain characteristic	High cost and larger in size

2. Performance Evaluation Criteria of Gain Equalizers

Four parameters exist that are mainly considered to evaluate the performance of Gain Equalizers: Slope Linearity, VSWR, Insertion Loss, and Slope Compensation Range.



Evaluation parameters of Gain Equalizers

Slope Linearity: Slope Linearity is defined as the deviation in dB between the measured curve and the Best Fit Straight Line as shown in Graph 6. Gain Equalizers whose measured curve deviates little from the Best Fit Straight Line are evaluated as superior. Gain Equalizers designed by professionals will provide linearity within ± 0.75 dB max. (± 0.5 dB typ.).

VSWR: VSWR is defined as the 50 Ohms matching characteristic of Gain Equalizers. Gain Equalizers designed by professionals will provide VSWR within 1.67:1 max. (1.5:1 typ.).

Insertion Loss: Gain Equalizers with little insertion loss at the highest frequency within their operating frequency range are evaluated as superior. Gain Equalizers designed by professionals will provide insertion loss within 1.75 dB max. (1.0 dB typ.).

Slope Compensation Range: Slope Compensation Range is defined as the range in dB in which Gain Equalizers compensate for gain characteristics. Gain Equalizers with more inclusive range are evaluated as better than others. Gain Equalizers designed by professionals will provide slope compensation range between 1 dB to 12 dB.

At 1 GHz, it has been this circuit is extremely difficult to construct and could not be made to give the required positive gain-frequency slope. But, with some sacrifice in return loss, the simpler, more practical equalizer topology of Fig. 3 can provide realistic results.

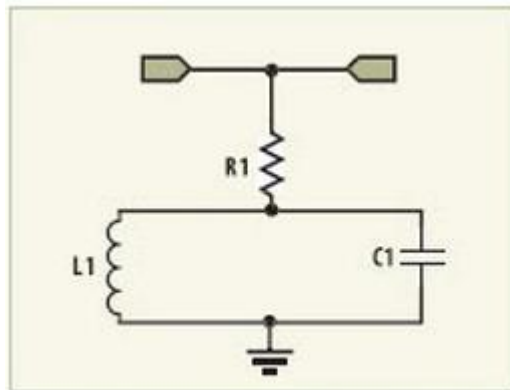


Fig. 1 Resistive Equalizer base on Lumped components

This provides a low component count, thereby reducing the effects of parasitic component pad capacitances and reducing the effects of non-ideal component properties. The equalizer essentially comprises a resistor in series with an LC resonator connected to ground. Again, the LC resonance is set to just above the frequency band of interest for the amplifier. The response well away from this resonance is governed by the resistor value and therefore higher values (typically greater than $110\ \Omega$) are recommended to maintain a reasonable wideband return loss. The slope may be adjusted by varying the LC component values (maintaining a similar resonance frequency) or by changing the resistor value. This topology has been built, measured, and successfully used to cancel the gain slope of a commercial MMIC amplifier over a 250-MHz bandwidth

Applications:

- Frequency Converter
- Multiband Cellular Phone Signal Repeaters
- Distribution Antenna Systems
- RF Transceiver Units

SET UP & PROCEDURE:

A. Using Spectrum Analyzer

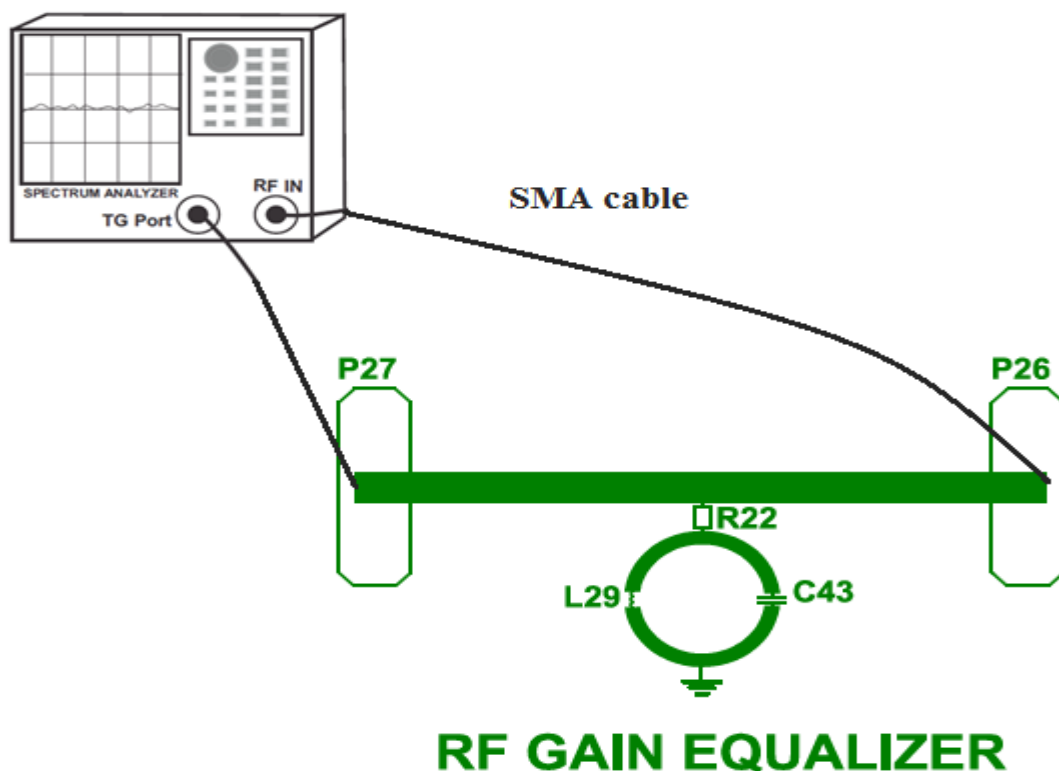
Measurement of Insertion Loss (S21) Characteristics of RF Gain Equalizer

Procedure.

Setting for Spectrum Analyzer

- Frequency Range : 100 MHz TO 1.0 GHz
- TG Source Power Level : 0 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.



Measurement of Insertion Loss characteristics (S21):

- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the Port 27 of the RF Gain Equalizer module (i.e. signal of the Spectrum analyzer is applied as input to the Gain Equalizer)
- 2) Connect the Port 26 of the RF Gain Equalizer to the Spectrum analyzer Port 2 (i.e. output from the Gain Equalizer is applied to the Spectrum analyzer for data display).

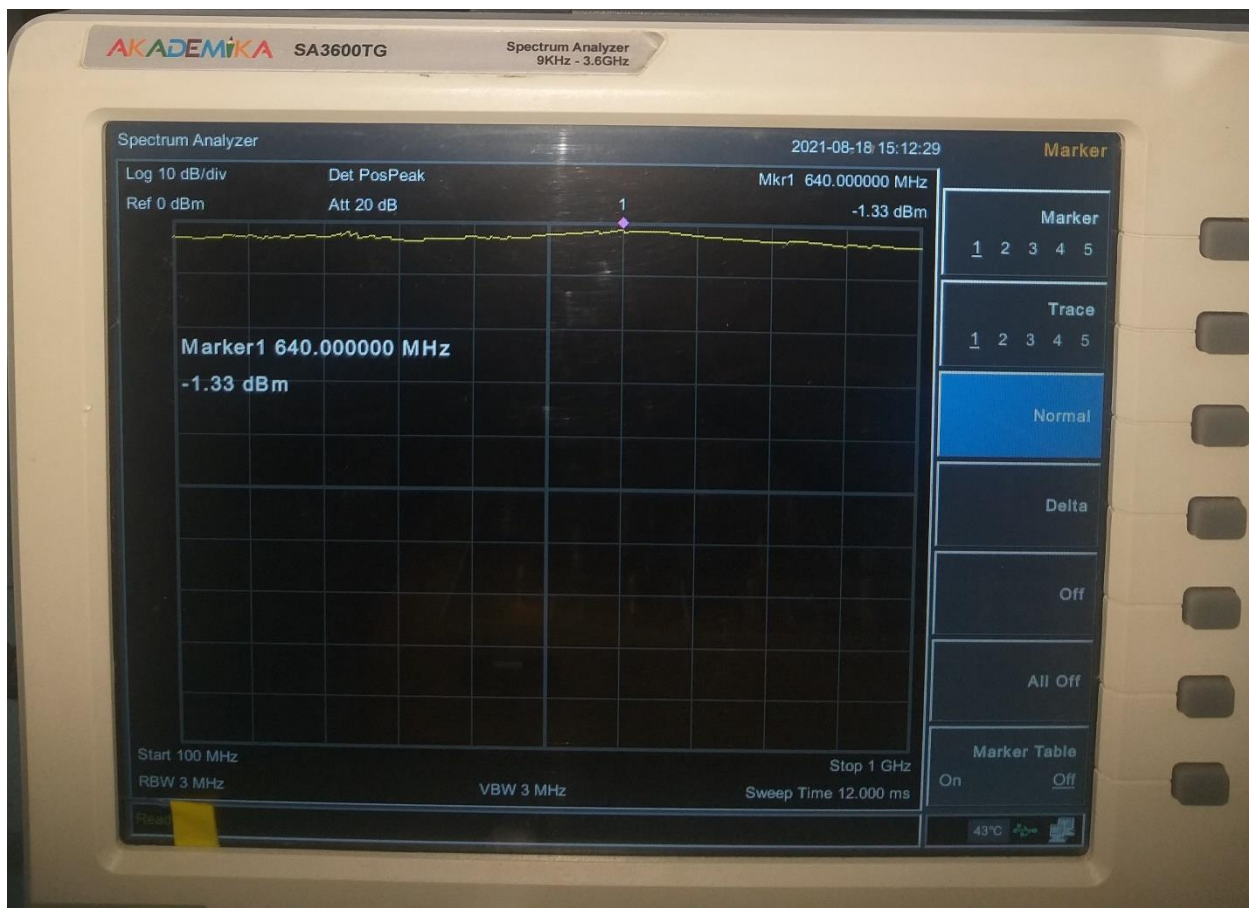
- 3) Apply markers and note down the readings for the frequencies specified in the observation table.
- 4) Observe the output of insertion Loss S21.

Observation Table

Sr.No	Freq (MHz)	Insertion Loss S21 (dB)
1.	600	
2.	620	
3.	640	
4.	660	
5.	700	

Summarize the insertion loss of Gain Equalizer for the frequencies specified in the observation table.

Reference Plot (S21):



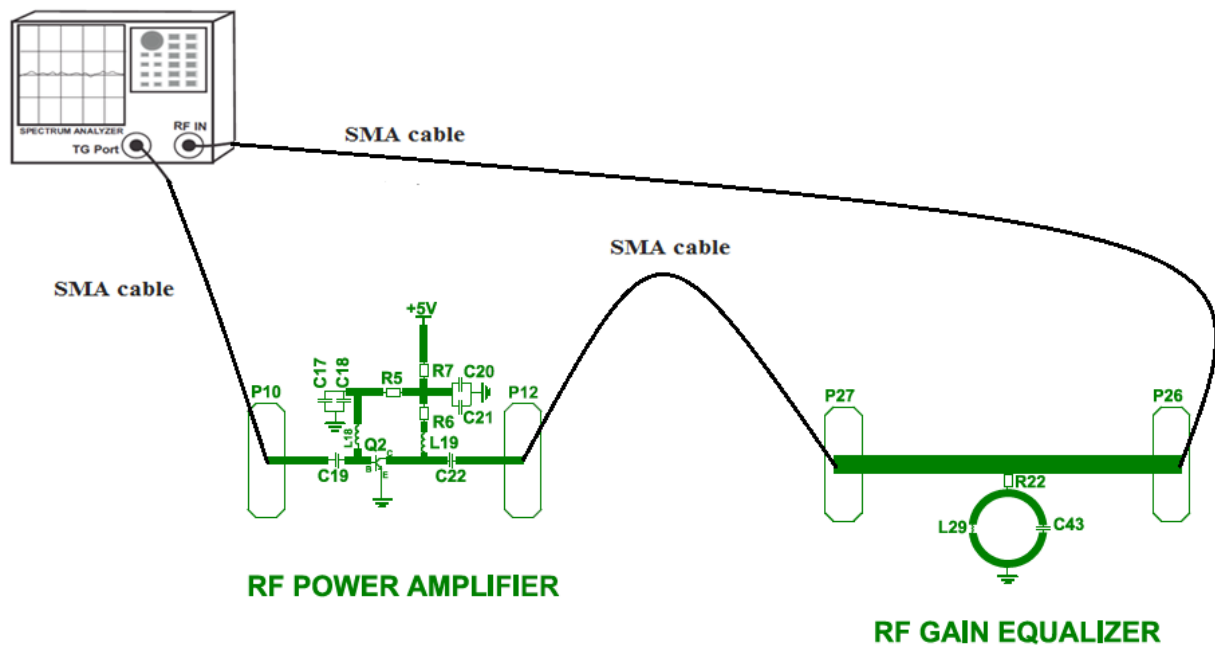
Power Amplifier Gain S21 Characteristics using Gain Equalizer

Procedure.

Setting for Spectrum Analyzer

- Frequency Range : 100 MHz TO 1.0 GHz
- TG Source Power Level : - 25 dBm

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.



Measurement of Power Amplifier Gain characteristics (S21):

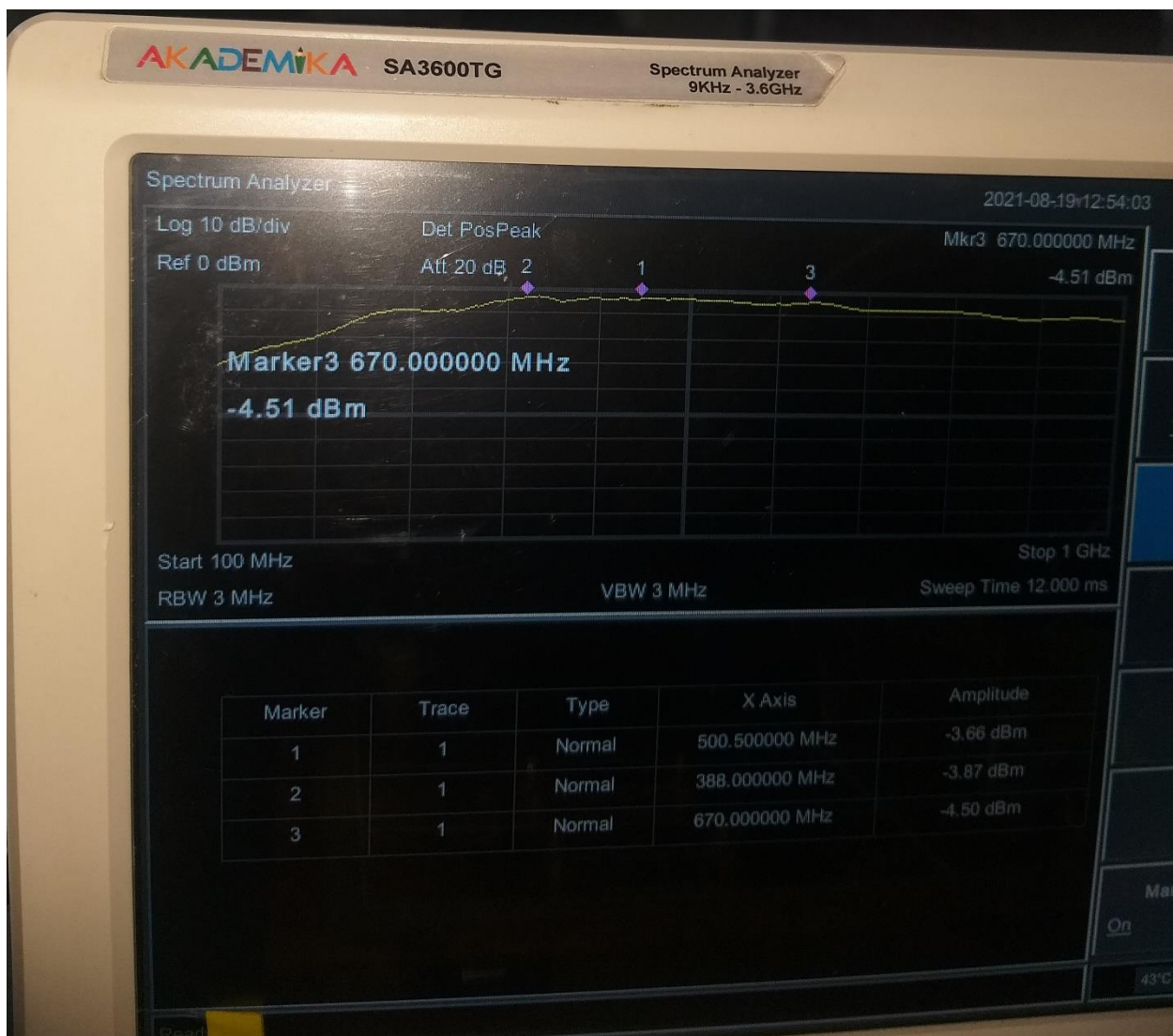
- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the Port 10 of the RF Power Amplifier module (i.e. signal of the Spectrum analyzer is applied as input to the Power Amplifier)
- 2) Connect the Power Amplifier output Port 12 to the RF Gain Equalizer input port P27.
- 3) Connect the Port 26 of the RF Gain Equalizer to the Spectrum analyzer Port 2 (i.e. output from the Gain Equalizer is applied to the Spectrum analyzer for data display).
- 4) Apply markers and note down the readings for the frequencies specified in the observation table.
- 5) Observe the output of Gain S21.

Observation Table

Sr.No	Freq (MHz)	Gain S21 (dB)
1.	388	
2.	500	
3.	670	

Summarize the Gain of RF Power Amplifier for the frequencies specified in the observation table.

Reference Plot Gain (S21):



Note : This is the reference graph plot. The graphs vary slightly according to RFT-A boards due to tolerance in PCB and components.

**EXPERIMENT
NO.6
RF LC OSCILLATOR**

EXPERIMENT NO: 6

Theory and Experimentation of RF LC Oscillator

OBJECTIVE

To comprehend the purpose and operational principle of the LC Oscillator by using the RFL-RFT-1G-A module

AIM

To explore the properties of the RF LC Oscillator module.

Study the LC Oscillator circuit

Measurement of Resonant Frequency

EQUIPMENTS

RFL-RFT-1G-A KIT, SPECTRUM ANALYZER

THEORY

A **Colpitts Oscillator** is a type of LC oscillator. Colpitts oscillators were invented by American engineer Edwin H. Colpitts in 1918. Like other LC oscillators, Colpitts oscillators use a combination of inductors (L) and capacitors (C) to produce an oscillation at a certain frequency. The distinguishing feature of the Colpitts oscillator is that the feedback for the active device is taken from a voltage divider made of two capacitors in series across the inductor.

Colpitts Oscillator Circuit

Other components in the circuit are the same as that found in the case of common-emitter CE, which is biased using a voltage divider network, i.e., R_C is the collector resistor, R_E is the emitter resistor which is used to stabilize the circuit, and the resistors R_1 and R_2 form the voltage divider bias network.

Figure 1 shows a typical Colpitts oscillator with a tank circuit. An inductor L is connected parallel to the serial combination of capacitors C_1 and C_2 (shown by the red enclosure).

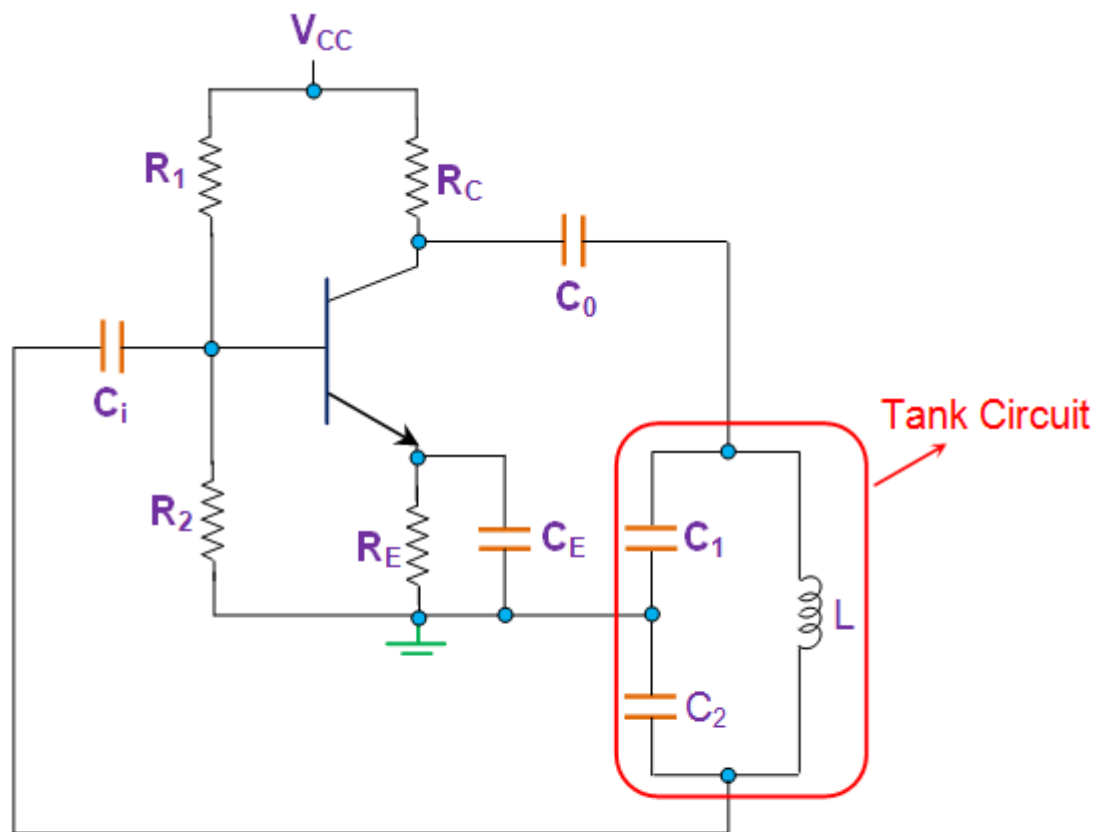


Figure 1 Colpitts Oscillator

Further, the capacitors C_i and C_o are the input and output decoupling capacitors while the emitter capacitor C_E is the bypass capacitor used to bypass the amplified AC signals. Here, as the power supply is switched ON, the transistor starts to conduct, increasing the collector current I_C due to which the capacitors C_1 and C_2 get charged. On acquiring the maximum charge feasible, they start to discharge via the inductor L .

During this process, the electrostatic energy stored in the capacitor gets converted into magnetic flux, which is stored within the inductor in the form of electromagnetic energy. Next, the inductor starts to discharge, which charges the capacitors once again. Likewise, the cycle continues, which gives rise to the oscillations in the tank circuit.

Further the figure shows that the output of the amplifier appears across C_1 and thus is in-phase with the tank circuit's voltage and makes-up for the energy lost by re-supplying it. On the other hand, the voltage feedback to the transistor is obtained across the capacitor C_2 , which means the feedback signal is out-of-phase with the voltage at the transistor by 180° .

This is due to the fact that the voltages developed across the capacitors C_1 and C_2 are opposite in polarity as the point where they join is grounded.

Further, this signal is provided with an additional phase-shift of 180° by the transistor which results in a net phase-shift of 360° around the loop, satisfying the phase-shift criterion of Barkhausen principle.

At this stage, the circuit can effectively act as an oscillator producing sustained oscillations by carefully monitoring the feedback ratio given by (C_1 / C_2) . The frequency of such a **Colpitts Oscillator** depends on the components in its tank circuit and is given by

$$F = \frac{1}{2\pi\sqrt{LC_{eff}}}$$

Where the C_{eff} is the effective capacitance of the capacitors expressed as

$$\frac{C_1 C_2}{C_1 + C_2}$$

As a result, these oscillators can be tuned either by varying their inductance or the capacitance. However, the variation of L does not yield a smooth variation. Hence they are usually tuned by varying the capacitances which are generally ganged, due to which a change in any one of them changes both of them. Nevertheless, the process is tedious and requires a special large-valued capacitor.

Thus, the Colpitts oscillators are seldom preferred in applications wherein the frequency varies but are more popular as fixed frequency oscillators due to their simple design.

SET UP & PROCEDURE:

A. Using Spectrum Analyzer

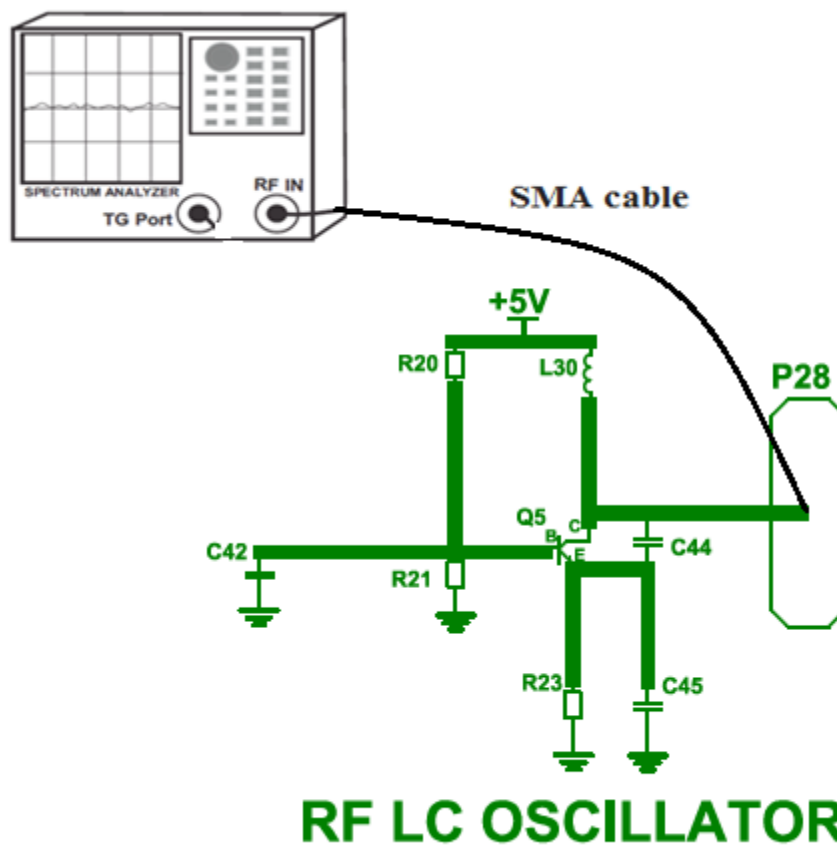
Frequency Measurement of LC Oscillators

Procedure.

Setting for Spectrum Analyzer

- Frequency Range : 20 MHz TO 40 MHz

IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.*



Setup for Frequency Measurement

Frequency Measurement:

- 1) Connect the Port 28 of the LC Oscillator to the Spectrum analyzer Port 2 (i.e. output from the LC Oscillator is applied to the Spectrum analyzer for data display).
- 2) Connect DC Power Supply to RFT-A Board.
- 3) Turn ON DC Power Supply.

- 4) The Frequency range of the Spectrum Analyser is set from 20 MHz to 40 MHz.
- 5) Apply markers and note down the readings for the frequencies specified in the observation table.
- 6) Observe the generating output frequency from LC oscillator.

Observation Table

Sr.No	Generating output frequency (MHz)	Output Power (dBm)
1.		

Summarize the generating output frequency from LC oscillator

Reference Plot :



Note : This is the reference graph plot. The graphs vary slightly according to RFT-A boards due to tolerance in PCB and component's.

**EXPERIMENT
NO.7
FREQUENCY
MULTIPLIER**

EXPERIMENT NO: 7

Theory and Experimentation of RF Frequency Multiplier

OBJECTIVE

To comprehend the purpose and operational principle of the RF Frequency Multiplier by using the RFL-RFT-1G-A module

AIM

To explore the properties of the RF Frequency Multiplier module.

Study the RF Frequency Multiplier circuit

Measurement of multiplication of frequency

EQUIPMENTS

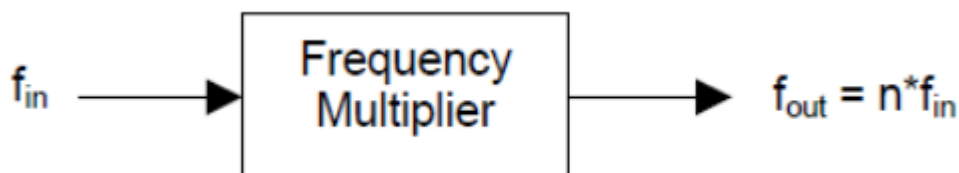
RFL-RFT-1G-A KIT, SPECTRUM ANALYZER

THEORY

Frequency Multiplier

In electronics, a frequency multiplier is an electronic circuit that generates an output signal whose output frequency is a harmonic (multiple) of its input frequency. Frequency multipliers consist of a nonlinear circuit that distorts the input signal and consequently generates harmonics of the input signal. A subsequent bandpass filter selects the desired harmonic frequency and removes the unwanted fundamental and other harmonics from the output.

A frequency multiplier has the property that the frequency of the output signal has an integer multiple of the input frequency.



Diode Frequency Multipliers Diode frequency multipliers can be generally classified as being of varistor (Schottky barrier diode) or varactor type. In the first case, frequency multiplication is performed by a nonlinear resistance or conductance with consequent poor conversion efficiency but a very large potential bandwidth. In the varactor type multiplier, a nonlinear reactive element (with nonlinear capacitance) is used. Varactor reactive type multipliers have high potential conversion efficiency (due to almost zero series resistance), needs very low drive power levels, but exhibit a narrow bandwidth and

a high sensitivity to operating conditions, and sometimes stability problems. Because the frequency multiplication is based on a nonlinear reactance, these circuits have tuned input and output circuits and often tuned idler circuits as well. In theory, a cascade of low-order multipliers usually has greater efficiency than a single high order multiplier, but must consider the additional losses in cascading two multipliers (it is invariably necessary to use an isolator between them), and especially the additional cost.

1. Resistive (Varistor) Frequency Multipliers

- Resistive frequency multipliers use the nonlinear I-V characteristic of a Schottky-barrier diode to distort a sinusoidal waveform. This distortion generates harmonics.
- The more is distorting the input sinusoid, the greater the harmonic currents in the diode, but the maximum still not very great because resistive frequency multipliers are not very efficient.
- In theory a diode doublers have 6dB conversion loss ($1/N^2$) but in reality, the conversion loss is about 10dB and there is no reason to make higher-harmonic resistive multipliers.
- The advantage of resistive multipliers is, they are very broadband

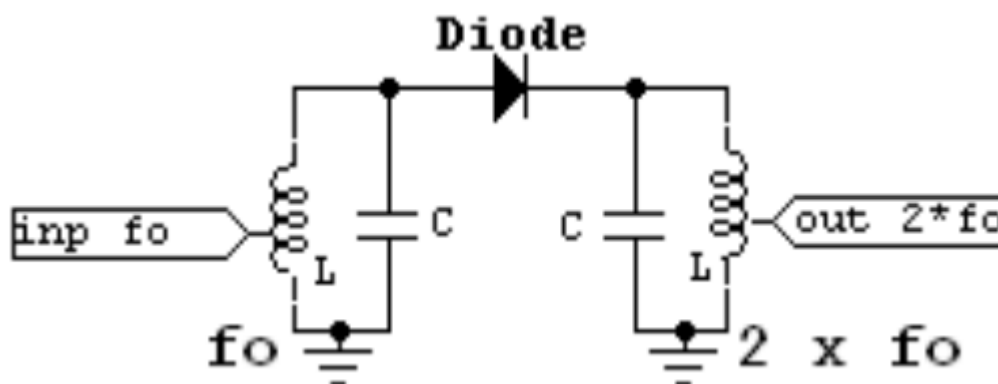


Fig.1 Simplified model of a Resistive (diode) Frequency Doubler

The parallel LC resonators are ideal because they short-circuit the diode at the unwanted harmonics, decoupling the input from the output, and put the diode in parallel with the input at the fundamental frequency and in parallel with the output at the second harmonic. The inductor can be tapped to optimize the source and load impedances of the diode.

2. Frequency doubler using microstrip lines

The frequency doubler using microstrip lines presented below is suitable for microwave frequencies.

The circuit uses a $\lambda/4$ at f_0 , short-circuited through a stub at the input side of the Schottky diode (which is equivalent to $\lambda/2$ at $2 \times f_0$), which is used to create a short-circuit at $2 \times f_0$ to prevent the output power generated in the diode from traveling backward. - Similarly use a $\lambda/4$ at f_0 open-ended stub at the output side, which creates an RF short at f_0 and causes the input signal penetrating through the diode to be reflected back to the diode. - A section of the transmission line is used as an inductor to resonate the diode junction capacitance.

- The $\lambda/4$ impedance transformers at the input and output are used to transform 50 ohms source and load to optimum diode impedance terminations

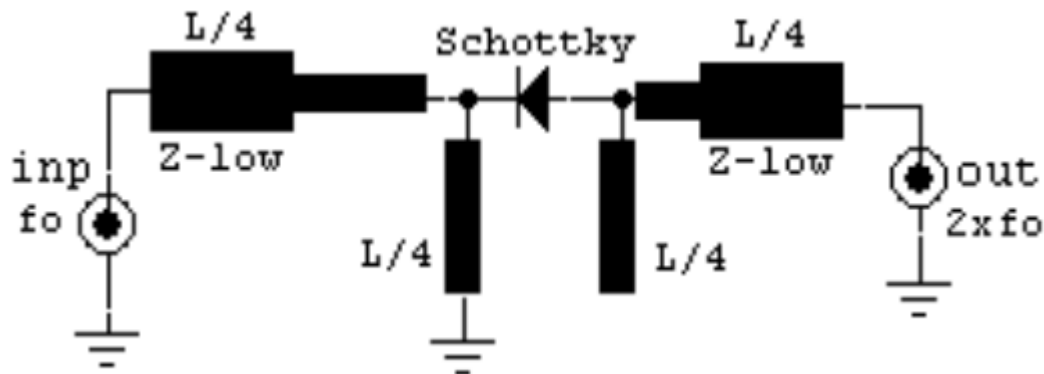


Fig.2 Microwave Microstrip Frequency Doubler

3. Nonlinear Transmission Line Multiplication

Frequency multiplication can also be accomplished with Nonlinear Transmission Lines (NLTL). They are similar to SRD only in the fact that they are both comb generators. Figure 3 is a distributed version using transmission lines and nonlinear capacitors

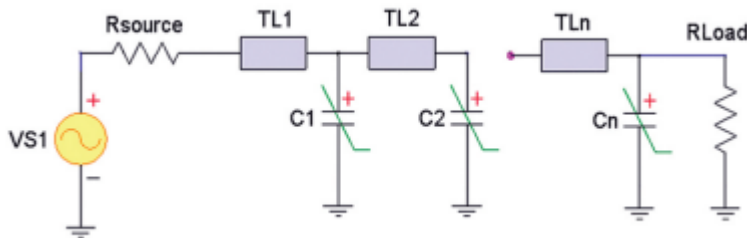


Figure 3: Distributed Nonlinear Transmission Line multiplier using nonlinear capacitors

Applications

- synthesizers
- local oscillators
- satellite up and down converters

SET UP & PROCEDURE:

Using Spectrum Analyzer

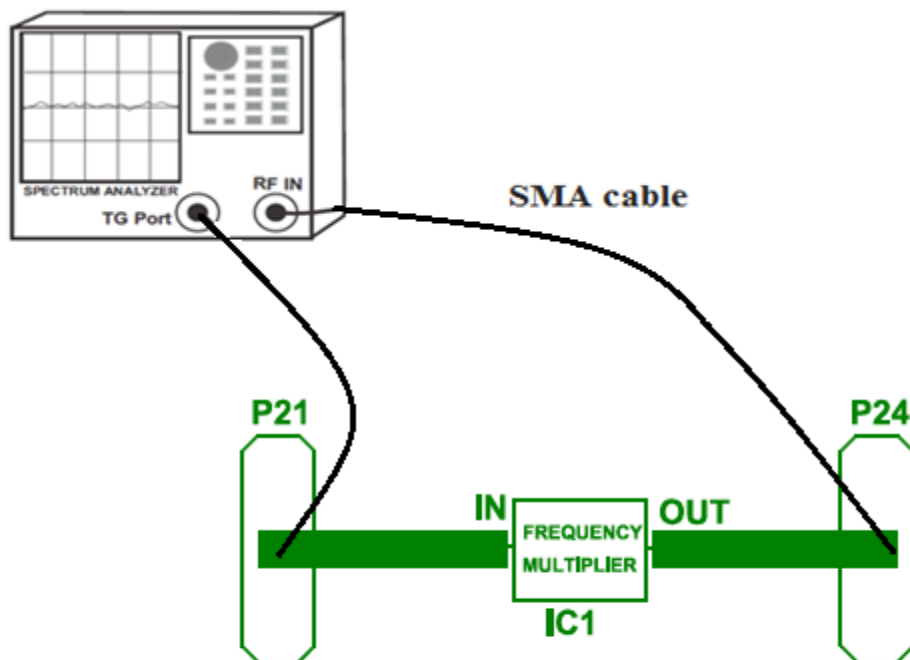
Output Frequency Measurement of Frequency Multiplier

Procedure.

Setting for Spectrum Analyzer

- Frequency Range : 40 MHz TO 150 MHz
- TG Source Power Level : 0 dBm
- TG output freq : 50 MHz

IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.*



RF FREQUENCY MULTIPLIER

Setup for Frequency Multiplication Measurement

Measurement of Frequency Multiplication

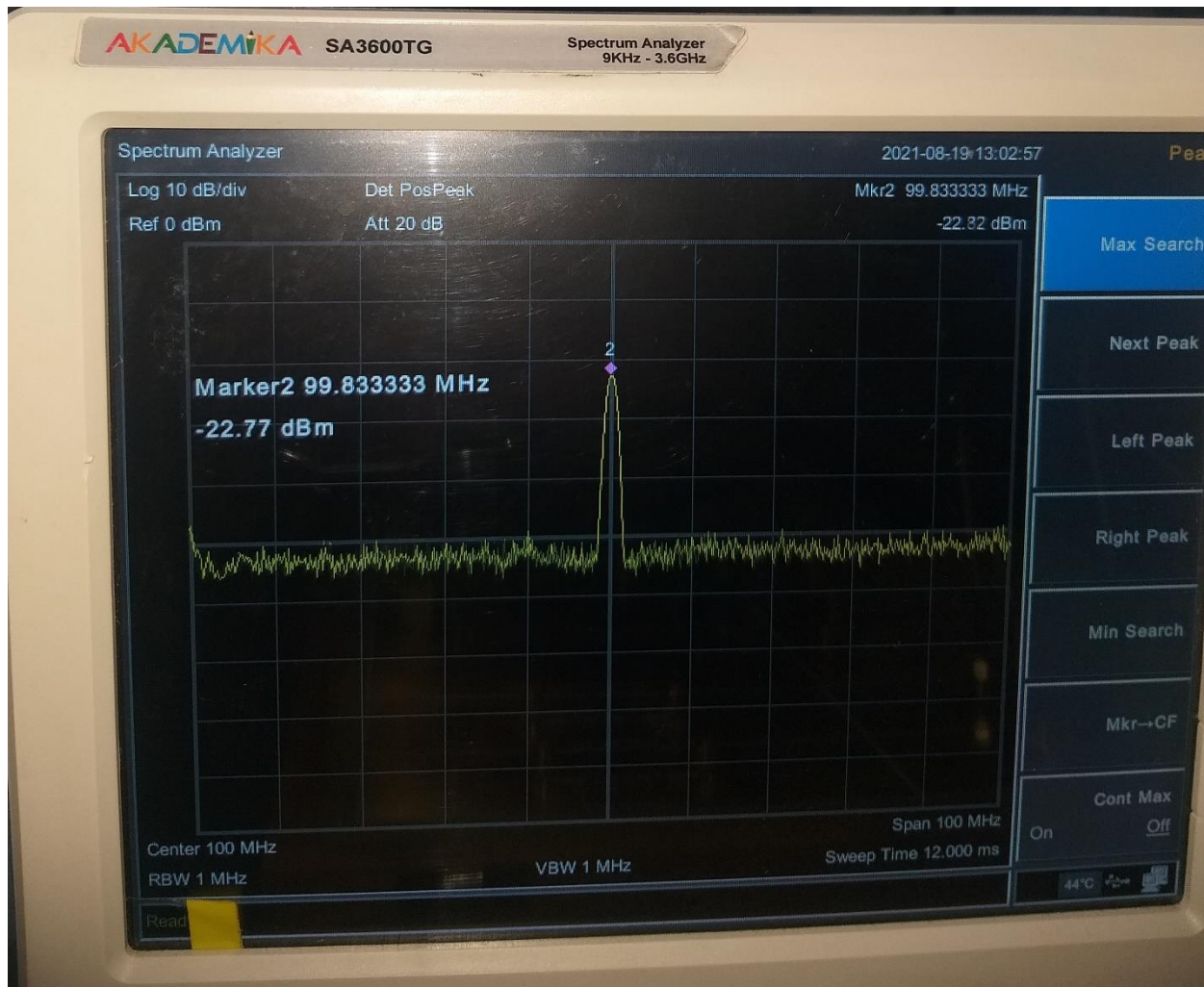
- 1) Connect the Tracking Generator (T.G. Port) output of the Spectrum analyzer to the Port 21 of the Frequency Multiplier module (i.e. signal of the Spectrum analyzer is applied as input to the Frequency Multiplier)
- 2) Connect the Port 24 of the Frequency Multiplier module to the Spectrum analyzer Port 2 (i.e. output from the Frequency Multiplier is applied to the Spectrum analyzer for data display).
- 3) Set the Tracking generation (TG) output of the Spectrum Analyzer to 0 dBm with TG freq set at 50 MHz - CW
- 4) The Frequency range of the Spectrum Analyser is set from 40 MHz to 150 MHz.
- 5) Apply peak search markers and note down the readings for the frequencies specified in the observation table.
- 6) Varied input frequency and observed output frequency as mention in table.

Observation Table

Sr.No	Input Frequency (MHz)	Output Frequency (MHz)
1.	50	
2.	55	
3.	60	
4.	65	
5.	70	

Summerize the input frequency and output frequency of multiplier

Reference Plot :



**EXPERIMENT
NO.8
PHASE-LOCKED
LOOP**

EXPERIMENT NO: 8

Theory and Experimentation of phase-locked loop

OBJECTIVE

To comprehend the purpose and operational principle of the phase-locked loop by using the RFL-RFT-1G-A module

AIM

To explore the properties of the phase-locked loop module.

Verify the Operation principle of PLL and study its performance.

EQUIPMENTS

RFL-RFT-1G-A KIT, Digital Storage Oscilloscope (DSO), Function generator (FG).

THEORY

A **phase-locked loop** or **phase lock loop (PLL)** is a control system that generates an output signal whose phase is related to the phase of an input signal. There are several different types; the simplest is an electronic circuit consisting of a variable frequency oscillator and a phase detector in a feedback loop. The oscillator generates a periodic signal, and the phase detector compares the phase of that signal with the phase of the input periodic signal, adjusting the oscillator to keep the phases matched.

Keeping the input and output phase in lock step also implies keeping the input and output frequencies the same. Consequently, in addition to synchronizing signals, a phase-locked loop can track an input frequency, or it can generate a frequency that is a multiple of the input frequency. These properties are used for computer clock synchronization, demodulation, and frequency synthesis.

Phase-locked loops are widely employed in radio, telecommunications, computers and other electronic applications. They can be used to demodulate a signal, recover a signal from a noisy communication channel, generate a stable frequency at multiples of an input frequency (frequency synthesis), or distribute precisely timed clock pulses in digital logic circuits such as microprocessors. Since a single integrated circuit can provide a complete phase-locked-loop building block, the technique is widely used in modern electronic devices, with output frequencies from a fraction of a hertz up to many gigahertz

Working Principle Of PLL

The phase-locked loop consists of a phase detector, a voltage controlled oscillator and, in between them, a low pass filter is fixed. The input signal ' V_i ' with an input frequency ' F_i ' is conceded by a phase detector. Basically the phase detector is a comparator that compares the input frequency f_i through the feedback frequency f_o . The output of the phase detector is $(f_i + f_o)$ which is a DC voltage. The out of the phase detector, i.e., DC voltage is input to the low pass filter (LPF); it removes the high-frequency noise and produces a steady DC level, i.e., $F_i - F_o$. The V_f is also a dynamic characteristic of the PLL.

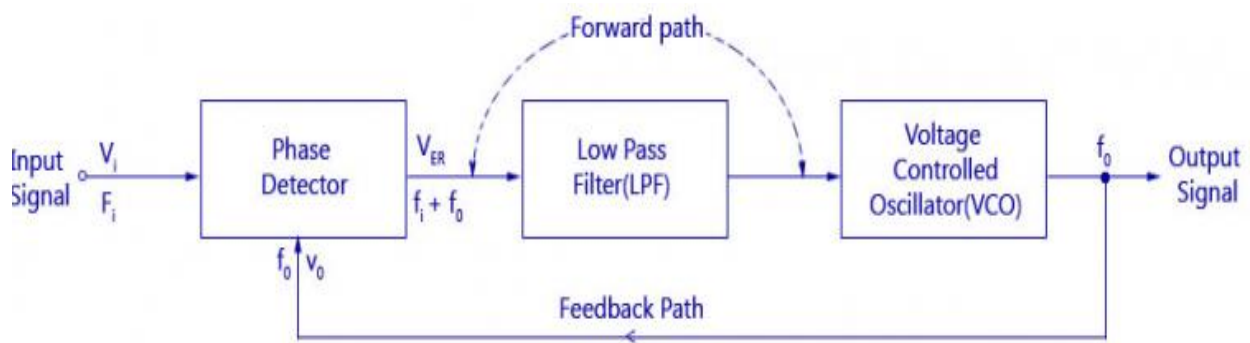


Fig.1 PLL Block Diagram

The output of the low pass filter, i.e., DC level is passed on to the VCO. The input signal is directly proportional to the output frequency of the VCO (f_o). The input and output frequencies are compared and adjusted through the feedback loop until the output frequency is equal to the input frequency. Hence, the PLL works like free running, capture, and phase lock.

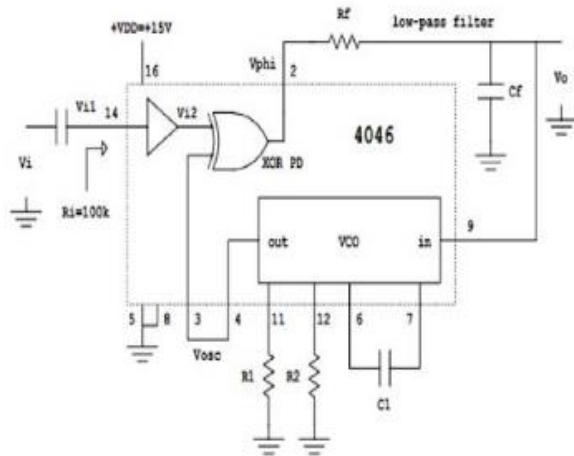
When there is no input voltage applied, then it is said to be a free-running stage. As soon as the input frequency applied to the VOC changes and produces an output frequency for comparison, it is called a capture stage. The below figure shows the block diagram of the PLL

Applications

Phase-locked loops are widely used for synchronization purposes; in space communications for coherent demodulation and threshold extension, bit synchronization, and symbol synchronization. Phase-locked loops can also be used to demodulate frequency-modulated signals. In radio transmitters, a PLL is used to synthesize new frequencies which are a multiple of a reference frequency, with the same stability as the reference frequency.

PLL DESIGN EXAMPLE

The filter output V_o controls the VCO, i.e., determines the frequency f_{osc} of the VCO output V_{osc} . From PLL 4046 circuit below, the voltage V_o controls the charging and discharging currents through capacitor C1. As a result the frequency f_{osc} of the VCO is determined by the V_o . The VCO output V_{osc} is a square wave with 50% duty ratio and frequency f_{osc} .



For proper operation of the VCO, components C1, R1 and R2 should satisfy: $100\text{pF} \leq C1 \leq 100\text{nF}$ and $10\text{k}\Omega \leq R1, R2 \leq 1\text{M}\Omega$

The VCO characteristics are adjustable by three components: R1, R2 and C1.

When $V_o = 0$, the VCO operates at the minimum frequency f_{min} given approximately by:

$$f_{min} = \frac{1}{R2(C1+32\text{ pF})}$$

When $V_o = V_{DD}$, the VCO operates at the maximum frequency f_{max} given approximately by:

$$f_{max} = f_{min} + \frac{1}{R1(C1+32\text{ pF})}$$

For $f_{min} \leq f_{osc} \leq f_{max}$, the VCO output frequency f_{osc} is ideally a linear function of the control voltage V_o .

The slope $K_o = \frac{\Delta f_{osc}}{\Delta V_o}$ of the $f_{osc}(V_o)$ characteristic is called the gain or the frequency sensitivity of the VCO, in Hz/V.

12

SET UP & PROCEDURE:

Using DSO and FG

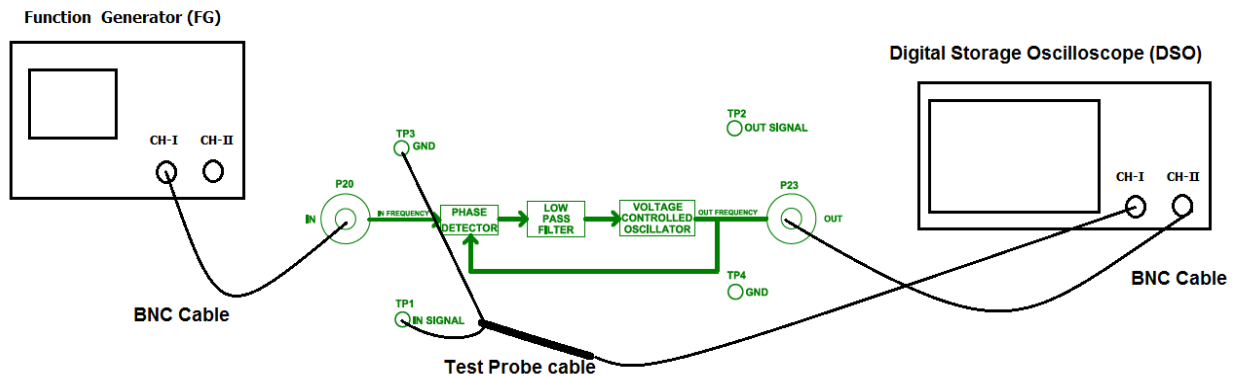
Study the input vs output frequency and phase of PLL circuit

Procedure.

Setting for Function Generator (FG)

- Input Frequency : 1.3 MHz
- Amplitude : 1 V

IMPORTANT: If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.



Setup connection diagram for PLL

- 1) Connect the power supply cables with proper polarity to RFT-A kit. While connecting this, ensure that the power supply is OFF.
- 2) Switch on the power supply.
- 3) Refer to the setup connection diagram and carry out the following connections.
- 4) Connect the 1.3 MHz frequency, 1Vpp signal from external function generator (Channel -1) to the IN of the PLL module (i.e. signal of the FG is applied as input to the PLL).
- 5) Connect the Test probe to the Test points of PLL IN (TP1, TP3) which is connected to the Channel-1 of DSO ((i.e. Input signal of the FG display on DSO).
- 6) Connect the OUT of the PLL to the Channel-2 of DSO (i.e. output signal of the PLL display on DSO).
- 7) Turn on Channel 2 and Observed input and output frequency of PLL on DSO.
- 8) Varied input frequency and observed output frequency as mention in table.
- 9) Varied input phase and observed output phase as mention in table.

Observation Table

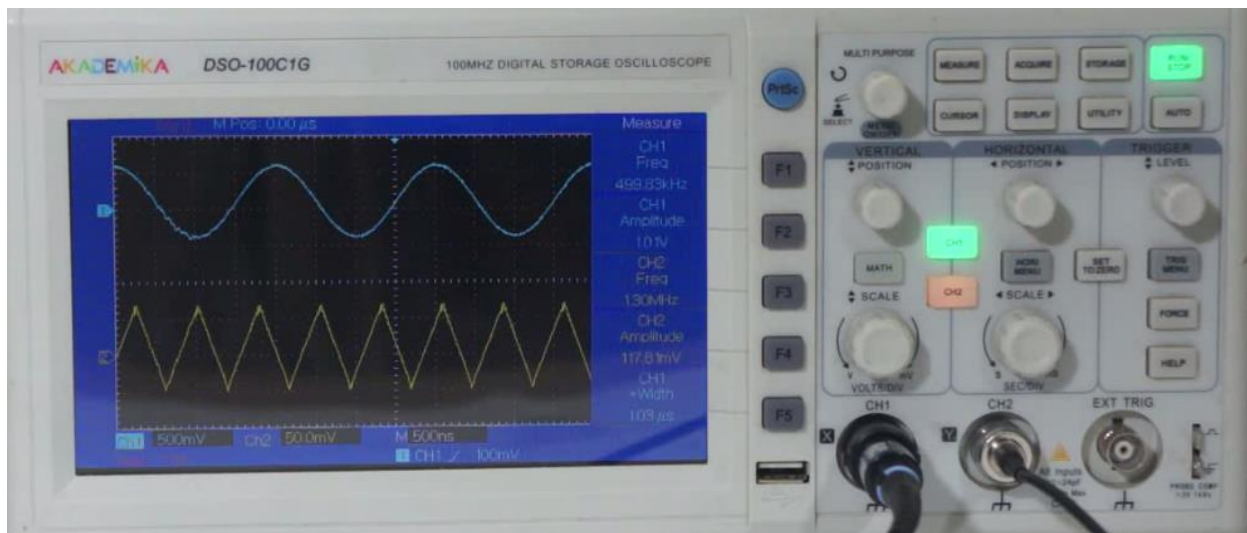
Sr.No	Input Frequency (MHz)	Output Frequency (MHz)
1.	0.5	
2.	1.3	
3.	2.0	
4.	3.0	

Sr.No	Input Frequency (MHz)	Input Phase (Degree)	Output Phase (IN /OUT Phase)
1.	1.3	0	
2.	1.3	180	

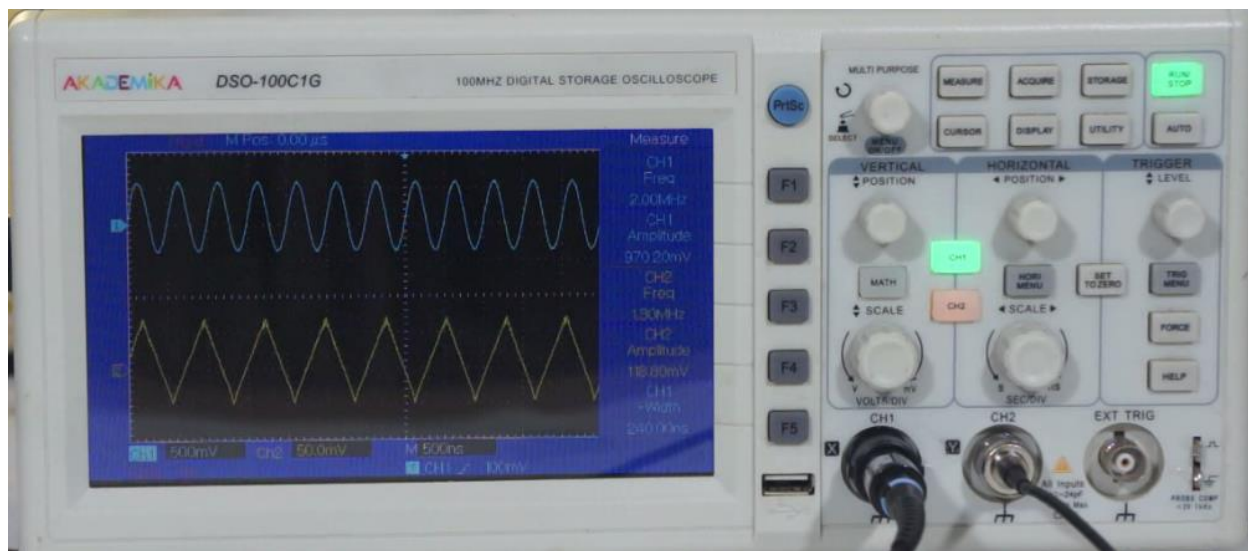
Summarize the input vs output frequency and phase of PLL.

Reference Plot for PLL :

1.For Input frequency 500 KHz



2. For Input frequency 2 MHz



Note : This is the reference graph plot. The graphs varying slightly according to RFT-A boards due to tolerance in PCB and component's.

EXPERIMENT NO.9 LUMPED FILTERS

EXPERIMENT NO: 9

Theory and Experimentation of Lumped Filters

OBJECTIVE

To comprehend the purpose and operational principle of the Lumped filters (LPF, HPF, BPF) by using the RFL-RFT-1G-A module.

AIM

- Filter characteristics like Measurement of insertion loss, -3dB Cut of freq of LPF,HPF and BPF.
- Study of configuration used for Lumped filter design.

EQUIPMENTS

RFL-RFT-1G-A KIT, SPECTRUM ANALYZER

THEORY

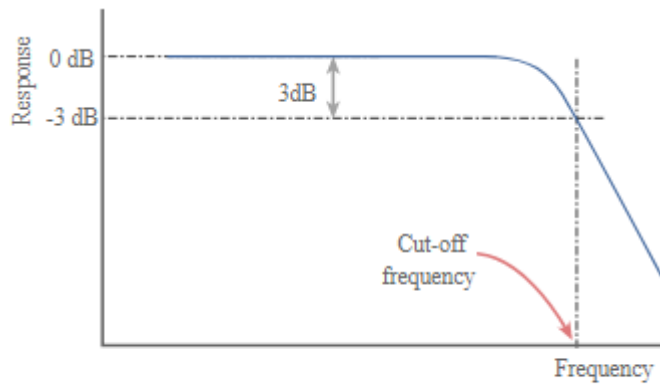
Filters are used in many areas of electronics. One of the main areas where they are used is within the radio frequency or RF domain. RF filters are used to remove or accept signals that fall in certain areas of the radio spectrum.

There are many different instances where they can be used - the list of applications is almost infinite. They are used within radio receivers to provide the selectivity, as well as only enabling the right band of frequencies to enter the latter parts of the set. They are used within transmitters to ensure that unwanted or spurious signals are not transmitted. RF filters are used to ensure that the required mix products from mixers are passed on to the next stages . . . the list of RF filter applications goes on.

Basic types of RF filter

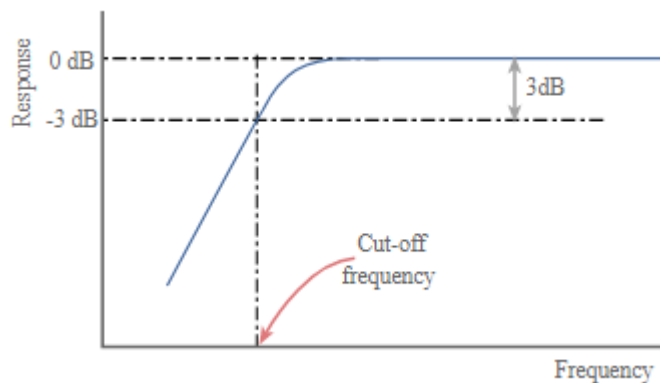
There are four types of filter that can be defined. Each different type rejects or accepts signals in a different way, and by using the correct type of RF filter it is possible to accept the required signals and reject those that are not wanted. The four basic types of RF filter are:

Low pass filter: As the name indicates the low pass filter is a form of filter that only allows through the lower frequencies. Typically it is nominally flat until the cut-off point, and then it rolls off.

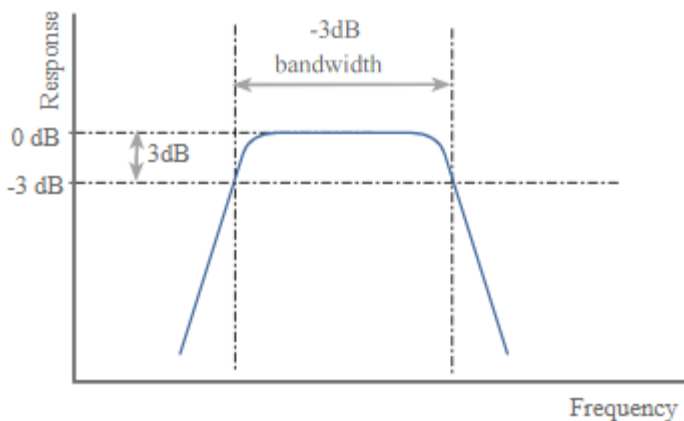


Generic low pass filter response

High pass filter: The high pass filter is in many ways the inverse of the low pass filter. It only allows signals through that are higher than the cut-off frequency. Above this point it is nominally flat, and below the RF filter cut-off frequency the response falls away at a rate determined by the order of the filter.



Band pass filter: The band pass RF filter only allows through signals within certain frequencies. Above and below the cut-off frequencies, the signals will be attenuated and within the accepted band of radio frequencies, signals will be passed through.



RF filter characteristics

- A filter allows signals through in what is termed the pass band. This is the band of frequencies below the cut off frequency for the filter.
- The cut off frequency of the filter is defined as the point at which the output level from the filter falls to 50% (-3 dB) of the in band level, assuming a constant input level. The cut off frequency is sometimes referred to as the half power or -3 dB frequency.

Filter classifications

Filters can be designed to meet a variety of requirements. Although using the same basic circuit configurations, the circuit values differ when the circuit is designed to meet different criteria. In band ripple, fastest transition to the ultimate roll off, highest out of band rejection are some of the criteria that result in different circuit values. These different filters are given names, each one being optimized for a different element of performance. Three common types of filter are given below.

- **Butterworth:** This type of filter provides the maximum in band flatness.
- **Bessel:** This filter provides the optimum in-band phase response and therefore also provides the best step response.
- **Chebyshev:** This filter provides fast roll off after the cut off frequency is reached. However this is at the expense of in band ripple. The more in band ripple that can be tolerated, the faster the roll off.
- **Elliptical:** This has significant levels of in band and out of band ripple, and as expected the higher the degree of ripple that can be tolerated, the steeper it reaches its ultimate roll off.

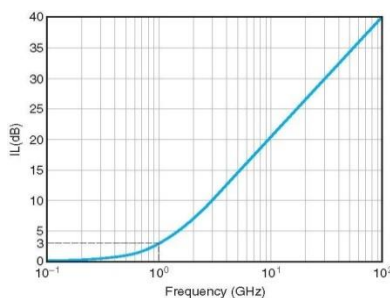


Fig: Butterworth

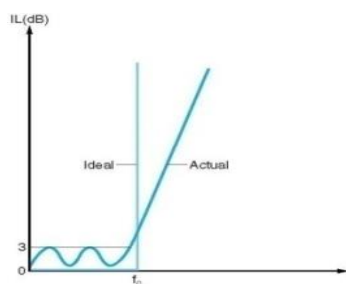


Fig: Chebyshev

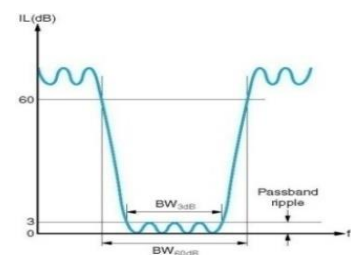
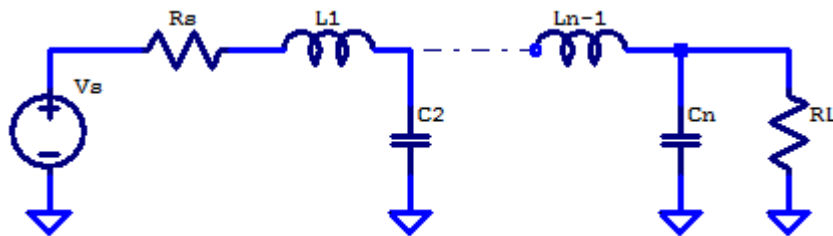


Fig: Elliptical

LC Butterworth Filter

Physical synthesis of a Butterworth filter requires components which can be interconnected to form complex poles (dampened harmonic oscillators). A passive arrangement of resistors and capacitors can form arbitrary real valued poles and zeros; however a passive RC network cannot synthesize complex poles. The natural choice for passive components which can synthesize oscillators, are inductors and capacitors. An immediate benefit of choosing an LC network for synthesis, is that the network in the ideal case is lossless. Allowing for low insertion-loss in the pass-band, and maximizing power transfer from source to load.

The most common network structure for synthesis is a series inductor and shunt capacitor ladder. A LC ladder of arbitrary order n , is shown in the figure below.



Butterworth filter design

This design really began by identifying and selecting the normalized approximated components coefficients using the standard coefficient table for Butterworth LPF.

Tabulated component values for prototype filters with 1 to 10 components is given in the table below

Table 1 : Component Values for Prototype Butterworth filter

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	2.0000	1.0000									
2	1.4142	1.4142	1.0000								
3	1.0000	2.0000	1.0000	1.0000							
4	0.7654	1.8478	1.8478	0.7654	1.0000						
5	0.6180	1.6180	2.0000	1.6180	0.6180	1.0000					
6	0.5176	1.4142	1.9318	1.9318	1.4142	0.5176	1.0000				
7	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	1.0000			
8	0.3902	1.1111	1.6629	1.9615	1.9615	1.6629	1.1111	0.3902	1.0000		
9	0.3473	1.0000	1.5321	1.8794	2.0000	1.8794	1.5321	1.0000	0.3473	1.0000	
10	0.3129	0.9080	1.4142	1.7820	1.9754	1.9754	1.7820	1.4142	0.9080	0.3129	1.0000

So in this case the coefficients are;

$$g_1 = 0.6180 \quad g_2 = 1.6180 \quad g_3 = 2.0000 \quad g_4 = 1.6180 \quad g_5 = 0.6180.$$

Using these coefficients the components values were computed and denormalized using the formulas (1), (2).

$$C_n = \frac{g_n}{2\pi F_C R} \dots\dots\dots (1)$$

$$L_n = \frac{g_n R}{2\pi F_C} \dots\dots\dots (2)$$

So for example;

$$C_1 = \frac{g_1}{2\pi F_C R}$$

$$C_1 = \frac{0.6180}{2\pi * 1\text{GHz} * 50}$$

$$C_n = 1.967\text{pF}$$

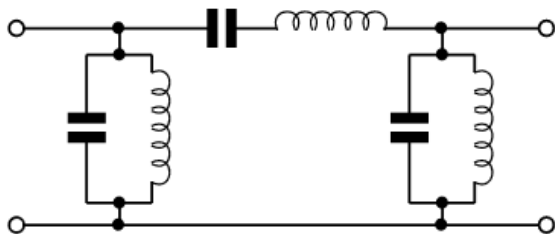
$$L_2 = \frac{g_2 R}{2\pi F_C}$$

$$L_2 = \frac{1.6180 * 50 g_{nR}}{2\pi * 1\text{GHz}}$$

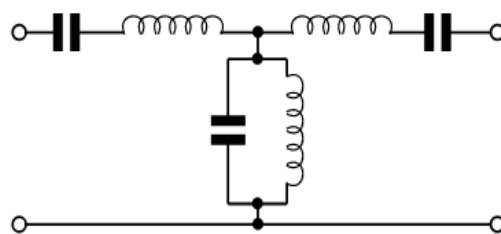
$$L_n = 12.876\text{nH}$$

LC band pass filter design

One of the easiest and most straight forward forms of filter to design is the constant-k filter. Like the high pass filters and the low pass filters, there are two topologies that are used for these filters, namely the Pi and the T configurations. Rather than having a single element in each leg of the filter as in the case of the low pass and high pass filters, the band pass filter has a resonant circuit in each leg. These resonant circuits are either series or parallel tuned LC circuits.

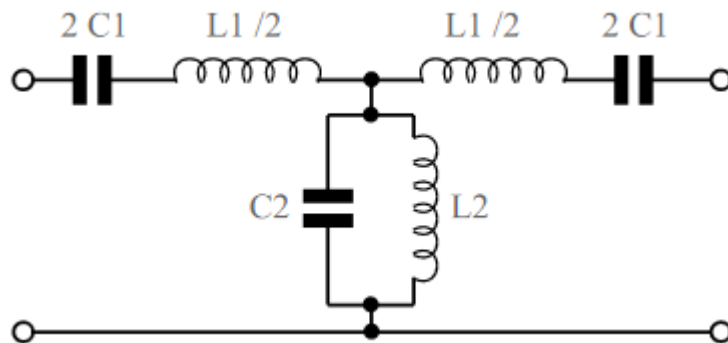


An example of a π format LC filter



An example of a T format LC filter

The equations below provide the values for the capacitors and resistors for a constant-k filter. As this constant-k filter is a band pass filter there are two cut off frequencies. One at the low edge of the pass band and the other at the top edge of the pass band.



T section band pass filter with values related to the equations

$$L_1 = \frac{Z_0}{\pi (f_2 - f_1)} \text{Henries}$$

$$L_2 = Z_0 \frac{f_2 - f_1}{4 \pi f_2 f_1} \text{Henries}$$

$$C_1 = \frac{f_2 - f_1}{4 \pi f_2 f_1 Z_0} \text{Farads}$$

$$C_2 = \frac{1}{\pi Z_0 (f_2 - f_1)} \text{Farads}$$

Where

Z_0 = characteristic impedance in ohms

C_1 and C_2 = Capacitance in Farads

L_1 and L_2 = Inductance in Henries

f_1 and f_2 = Cut off frequencies in Hertz

SET UP & PROCEDURE:

A. Using Spectrum Analyzer

Measurement of Insertion Loss (S21) Characteristics of Lumped Filters

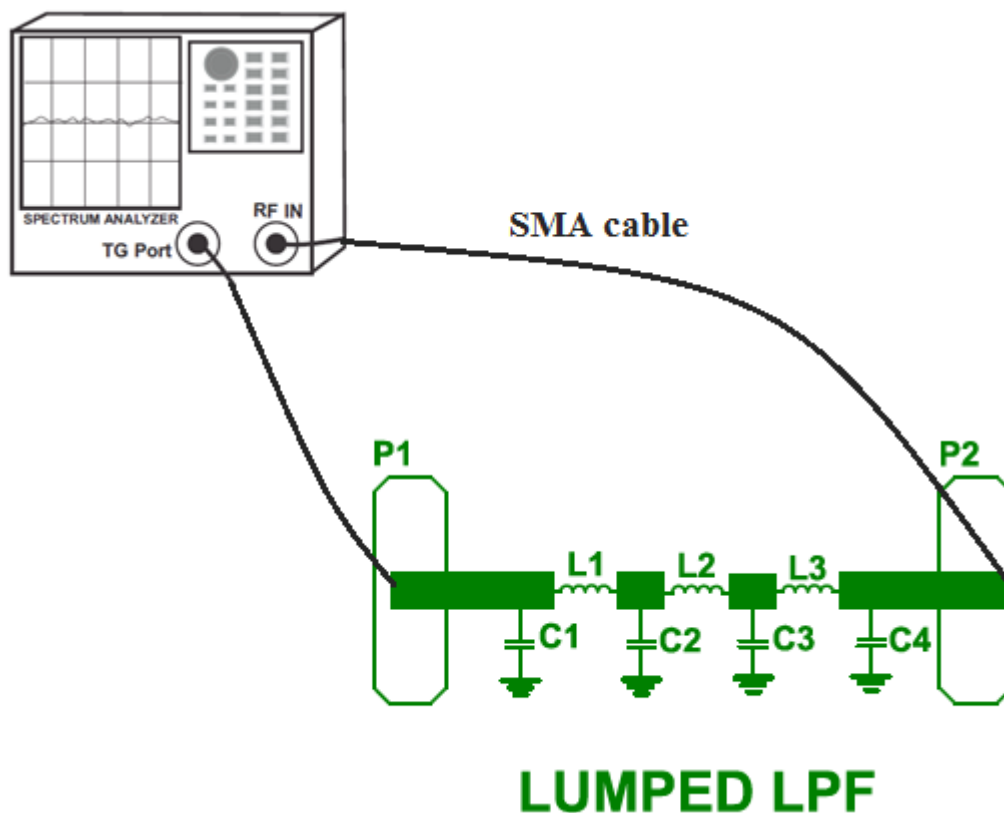
Procedure.

Setting for Spectrum Analyzer

- Frequency Range : 10 MHz To 1.0 GHz
- TG Source Power Level : 0 dBm

IMPORTANT: *If for some reason any of the above settings are changed while performing the experiment, the data displayed may not be reliable.*

Measurement of Insertion Loss characteristics for LPF (S21):



Setup for Insertion Loss (S21) Measurement

PROCEDURE: For observing the Passband Characteristics of the Lumped LPF module

- 1) Connect the **Tracking Generator** (T.G. Port) output of the Spectrum analyzer to the port 1 of the Lumped LPF module (i.e. signal from the spectrum Analyzer is applied as input to the LPF)
- 2) Connect Port 2 of the Lumped LPF module to the Spectrum analyzer **RF-IN Port** (i.e. output from the LPF is applied to the spectrum Analyzer for data display)
- 3) The displayed trace on the Spectrum Analyzer is of Insertion loss (S21) for the signal measured at Port 2 and RF input at Port 1.
- 4) Apply markers and take the reading for the frequencies mentioned in the observation table to determine the pass band insertion loss, 3 dB cut-off frequency.

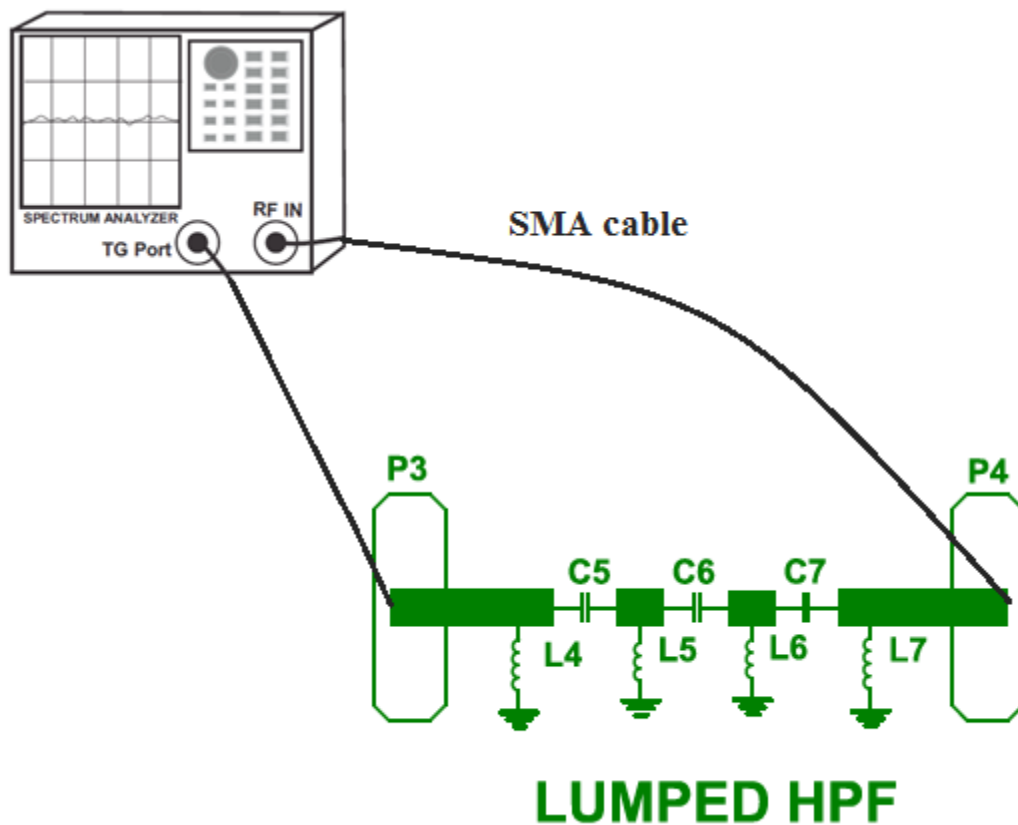
OBSERVATION TABLE:

Frequency (MHz)	Insertion loss S21) (dB)
100	
200	
300	
400	
440	
500	
550	
600	
700	

Summarize the Passband Insertion loss, 3dB cutoff frequency of Lumped LPF.

Reference Plot for Lumped LPF (S21) :

Measurement of Insertion Loss characteristics for Lumped HPF (S21):



Setup for Insertion Loss (S21) Measurement

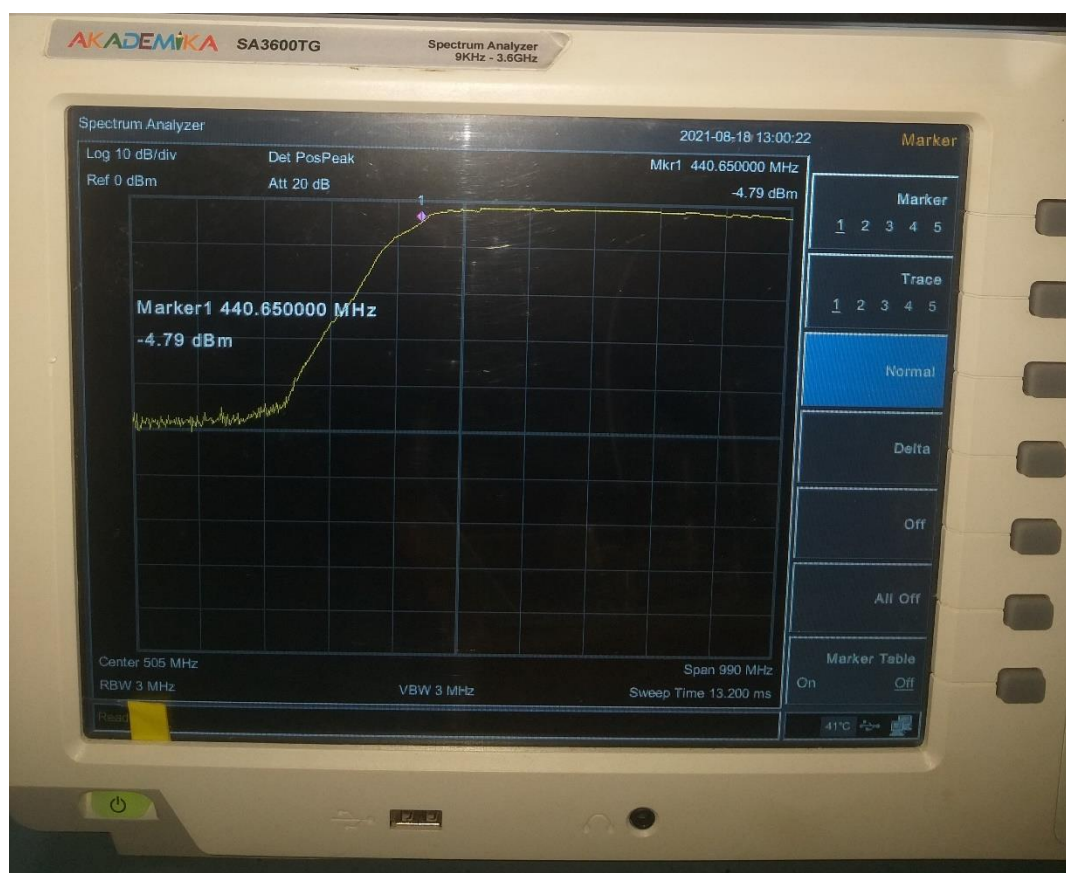
PROCEDURE: For observing the Passband Characteristics of the Lumped HPF module

- 1) Connect the **Tracking Generator** (T.G. Port) output of the Spectrum analyzer to the port 3 of the Lumped HPF module (i.e. signal from the spectrum Analyzer is applied as input to the HPF)
- 2) Connect Port 4 of the Lumped HPF module to the Spectrum analyzer **RF-IN Port** (i.e. output from the HPF is applied to the spectrum Analyzer for data display)
- 3) The displayed trace on the Spectrum Analyzer is of Insertion loss (S21) for the signal measured at Port 2 and RF input at Port 1.
- 4) Apply markers and take the reading for the frequencies mentioned in the observation table to determine the pass band insertion loss, 3 dB cut-off frequency.

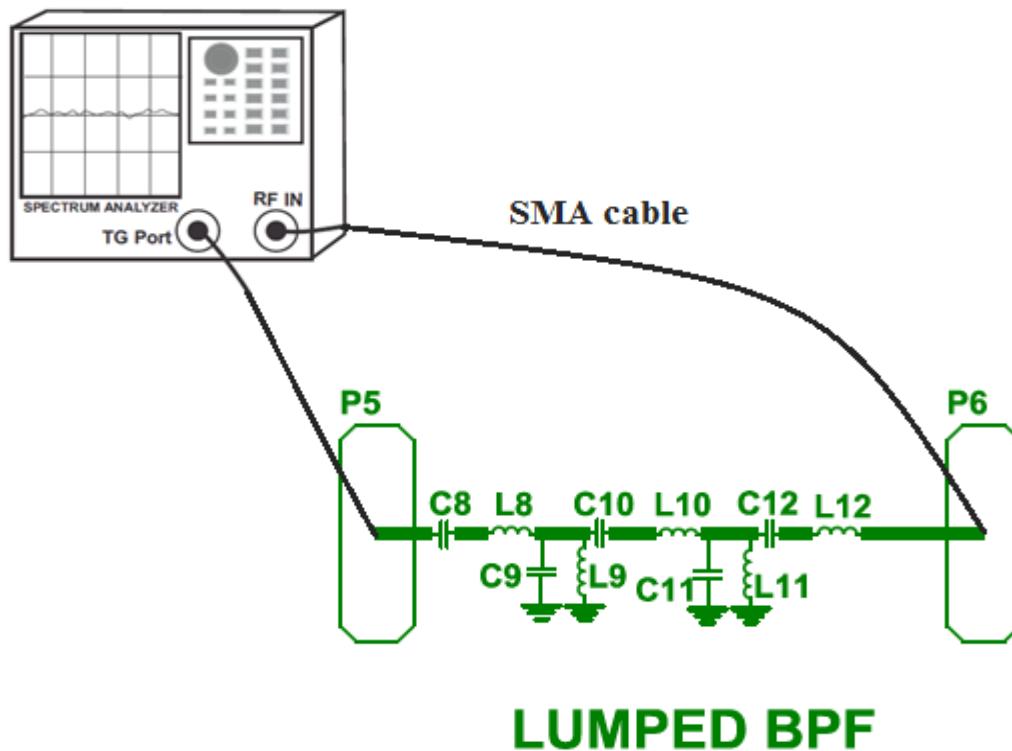
OBSERVATION TABLE:

Frequency (MHz)	Insertion loss S21 (dB)
400	
440	
500	
600	
700	
800	
900	

Summarize the Passband Insertion loss, 3dB cutoff frequency of lumped HPF.

Reference Plot for Lumped HPF (S21) :

Measurement of Insertion Loss characteristics Lumped BPF (S21):



Setup for Insertion Loss (S21) Measurement

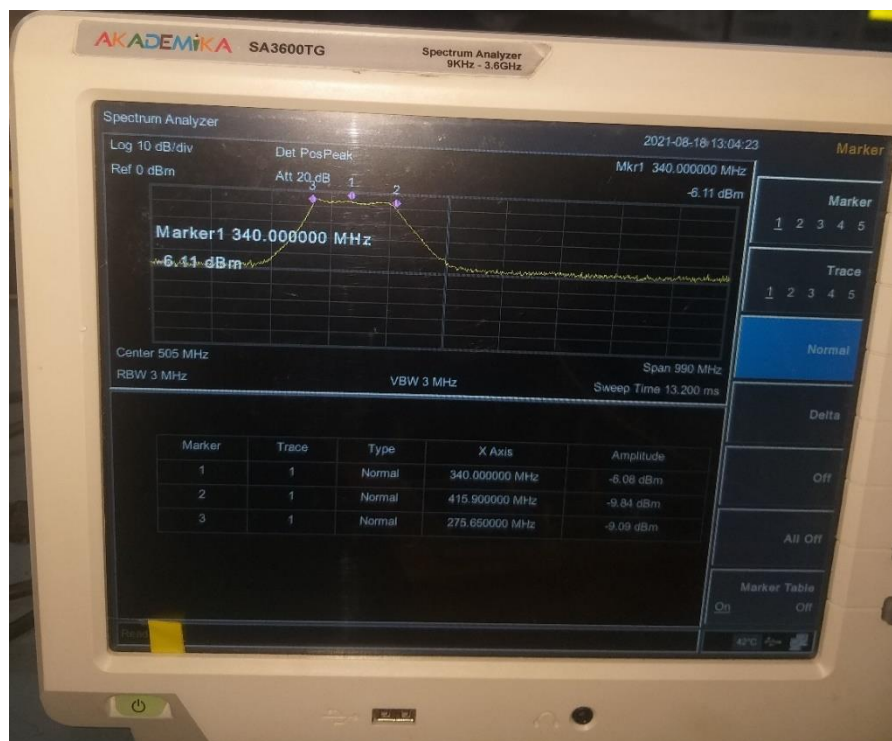
PROCEDURE: For observing the Passband Characteristics of the Lumped BPF module

- 1) Connect the **Tracking Generator** (T.G. Port) output of the Spectrum analyzer to the port 5 of the Lumped BPF module (i.e. signal from the spectrum Analyzer is applied as input to the BPF)
- 2) Connect Port 5 of the Lumped BPF module to the Spectrum analyzer **RF-IN Port** (i.e. output from the BPF is applied to the spectrum Analyzer for data display)
- 3) The displayed trace on the Spectrum Analyzer is of Insertion loss (S21) for the signal measured at Port 2 and RF input at Port 1.
- 4) Apply markers and take the reading for the frequencies mentioned in the observation table to determine the pass band insertion loss, 3 dB cut-off bandwidth of Lumped BPF.

OBSERVATION TABLE:

Frequency (MHz)	Insertion loss S21 (dB)
200	
250	
340	
415	
560	
650	

Summarize the Passband Insertion loss, 3dB and the 15dB bandwidth of lumped BPF.

Reference Plot for Lumped BPF (S21) :

Note : This is the reference graph plot. The graphs vary slightly according to RFT-A boards due to tolerance in PCB and component's.